Dynamic Vacuum Coupling Hypothesis (DVCH): A UV-Regularized Interacting-Vacuum Framework with Background Closure, Global Stability, and Diagnostic Tests

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We present a formulation of the *Dynamic Vacuum Coupling Hypothesis* (DVCH), a nonlinear interacting-vacuum scenario treated as an effective macroscopic closure within a covariant interacting dark-sector framework. The model combines power-law tracking between vacuum and matter densities with an ultraviolet (UV) curvature suppression inspired by higher-curvature effective field theory (EFT). We (i) provide a consistent covariant action-based motivation for dark-sector interaction via conformal matter coupling to a scalar EFT, (ii) derive the exact implicit background system in terms of $E(z) = H(z)/H_0$ and obtain the exact energy-transfer function Q implied by the DVCH closure, (iii) show that the effective vacuum equation of state satisfies $w_{\Lambda,\mathrm{eff}} > -1$ whenever $Q < 0$, while any numerical phantom-divide crossing must correspond to a sign change of Q or to an inconsistent plotting convention, (iv) derive the conditions under which the autonomous background system approaches a Minkowski end state, and (v) confront the model with real cosmological data at background level: a Markov Chain Monte Carlo analysis of the Pantheon+ supernova sample restricted to $z > 0.01$ and to the non-calibrator subsample (1580 SNe), the DESI DR1 BAO measurements (12 correlated D_M/r_d and D_H/r_d observables plus the two isotropic D_V/r_d points), and a 31-point cosmic-chronometer compilation, with the sound horizon r_d marginalised rather than fixed to the Planck value. On these data DVCH improves the best fit by $\Delta\chi^2 = -3.6$ over ΛCDM with two extra parameters, i.e. $\Delta\mathrm{AIC} = +0.4$: *no significant* preference, and the interacting model is statistically equivalent to the w_0w_aCDM control. The DVCH tracking index is constrained to $n = 0.52^{+0.10}_{-0.17}$ with $\beta = 1.10^{+0.63}_{-0.72}$. A Cobaya pipeline with a DVCH-patched CAMB against the Planck 2018 TT/TE/EE likelihood was executed, but we report it as provisional and not yet a comparison of the DVCH closure itself: the CAMB patch leaves the background identical to ΛCDM (verified numerically), so only the CDM perturbation term carries the new physics, and the nuisance sector, convergence and minimisation steps fall short of a referee-grade analysis. In this form, DVCH is presented as an effective interacting-vacuum framework whose background-level phenomenology is *consistent* with the expansion history but not favoured over ΛCDM at cosmological-precision level, while full perturbative and joint CMB consistency remains open.

I. INTRODUCTION

The standard ΛCDM model, while remarkably successful, is under intense scrutiny because of the persistent "Hubble tension"—a 5σ discrepancy between early-universe measurements from Planck and local late-time observations from SH0ES [1, 2]. In addition, the S_8 tension suggests that the Universe is less "clumpy" than predicted by the baseline model [3, 4].

Interacting Dark Energy (IDE) models, including the Dynamic Vacuum Coupling Hypothesis (DVCH) developed here, provide a useful laboratory for testing whether nontrivial energy-momentum exchange Q between dark sectors can remain both theoretically consistent and observationally viable [5–7]. Unlike ad hoc parameterizations that can trigger early-time instabilities, DVCH adopts a macroscopic closure motivated by ultraviolet (UV) curvature regularization and leads to a qualitatively distinct late-time Minkowski attractor. Here the goal is to define an internally consistent interacting-vacuum closure and to document background, stability, growth, and likelihood diagnostics, and to confront the closure with real Pantheon+/DESI DR1/cosmic-chronometer data and with the Planck 2018 high-ℓ likelihood [8–10].

This work develops DVCH as a *macroscopic closure* for a covariant interacting dark sector. The guiding philosophy is explicit: we do not claim uniqueness of a UV-complete microphysical realization, but we do require internal consistency, correct conservation-law bookkeeping, stable perturbative prescriptions, and clear numerical implementability.

Validation framework for DVCH

This work records a set of consistency checks for DVCH models, combining the covariant action-based motivation, the macroscopic background closure, the exact interaction kernel, the autonomous-system stability analysis, and the structure-growth diagnostic. The components treated here are the covariant motivation, the background closure, the exact interaction kernel, the global stability analysis, the sub-horizon growth calculation, the (n, β) background robustness scan, a real-data late-time

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likelihood analysis, and a confrontation of the closure against the real Planck 2018 likelihood with a DVCH-patched CAMB in Cobaya. The repository also includes a passing suite of 24 automated numerical regression tests for these local calculations. The suite checks normalization, finite and positive background solutions, interaction and growth outputs, likelihood behavior, MCMC diagnostic formulas, source-table generation, error handling, preflight reporting, the corrected real-data sets (DESI DR1, cosmic chronometers), agreement of the fitted DVCH background with the reference provider, and the presence of generated MCMC diagnostics. It is a local regression suite for the consistency of the implementation itself.

The CMB/Planck requirements are represented explicitly in the diagnostic package: a DVCH background/source-table interface for CLASS/CAMB-style backend checks [11, 12], the photon–baryon/CMB component checklist, angular-spectrum and lensing targets, a Planck `clik` likelihood configuration target [13], and a Cobaya-compatible Planck-stack configuration target [14]. Of these, the patched-CAMB runtime together with the official Planck likelihood and data *has* been executed, but the run is provisional: the CAMB patch leaves the background identical to ΛCDM, the chain stopped at $R-1=0.03$, and the free-nuisance re-optimised minima (though now computed) measure only the CDM perturbation term, as reported in Sec. IX D; the remaining components (a DVCH background patch, and a photon–baryon hierarchy with the interaction) are still external integration targets, and the perturbative CMB validation claim is reserved for that future runtime. The aim is to state the assumptions and limitations of the UV-regularized interacting-vacuum closure and to separate local calculations from CMB literature benchmarks.

II. COVARIANT THEORETICAL BASIS: AN INTERACTING DARK SECTOR FROM AN EFT-MOTIVATED ACTION

A frequent referee objection to IDE phenomenology is the absence of a covariant field-theory structure explaining how an interaction Q arises. A minimal and standard approach is to embed the interaction in a scalar-tensor EFT where cold matter couples conformally to the scalar while radiation remains minimally coupled. This produces an *unambiguous* covariant energy-exchange vector Q^μ.

A. EFT-inspired action (covariant, minimal)

We consider (in units $\hbar=c=1$ unless explicitly restored)

$$S=\int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R-\frac{1}{2}(\nabla\phi)^2-V(\phi)+\frac{\gamma}{2}\ell_{\rm P}^2R^2\right] + S_m\big[\psi_m,\,A^2(\phi)\,g_{\mu\nu}\big]+S_r[\psi_r,\,g_{\mu\nu}]\,.$$

Here $A(\phi)$ is a conformal coupling function. The R^2 term is the simplest local higher-curvature EFT correction, in the same broad class of curvature corrections used in early-universe model building [15]; its role is *motivational* for a high-curvature suppression scale and does not dominate at observable redshifts.

Varying the action yields

$$\nabla_\mu T^{\mu\nu}_{(m)}=+Q^\nu,\qquad \nabla_\mu T^{\mu\nu}_{(\phi)}=-Q^\nu,\qquad \nabla_\mu T^{\mu\nu}_{(r)}=0, \tag{1}$$

with the standard scalar-tensor exchange vector

$$Q^\nu=\left(\frac{\mathrm{d}\ln A}{\mathrm{d}\phi}\right)T_{(m)}\,\nabla^\nu\phi,\qquad T_{(m)}\equiv T^{\mu}{}_{(m)\mu}\simeq-\rho_m. \tag{2}$$

In FLRW, $Q^\nu=Qu^\nu$ with

$$Q=-\left(\frac{\mathrm{d}\ln A}{\mathrm{d}\phi}\right)\rho_m\dot\phi. \tag{3}$$

Equation (3) provides a *covariant* origin for energy transfer. DVCH will now be formulated as a *closure relation* for ρ_Λ (interpreted as the effective vacuum-like dark energy), which in turn determines Q at the background level.

B. Macroscopic closure vs. microscopic model

At the level of homogeneous cosmology, many microphysical models map into an effective two-fluid system:

$$\dot\rho_r+4H\rho_r=0, \tag{4}$$
$$\dot\rho_m+3H\rho_m=-Q, \tag{5}$$
$$\dot\rho_\Lambda=Q, \tag{6}$$

where ρ_Λ is vacuum-like in the sense of a baseline pressure $p_\Lambda=-\rho_\Lambda$, while the interaction induces an effective equation of state (Sec. IV).

DVCH is the choice of a specific functional relation $\rho_\Lambda=\rho_\Lambda(\rho_m,H)$ motivated by tracking and curvature suppression; given this closure, Q is fixed by (6). This replaces ambiguous ad hoc $Q\propto H\rho_i$ ansätze by an explicit Q computed from a stated $\rho_\Lambda(\rho_m,H)$.

III. DVCH CLOSURE AND IMPLICIT BACKGROUND COSMOLOGY

We assume flat FLRW with $H\equiv\dot a/a$ and

$$3M_{\rm Pl}^2H^2=\rho_r+\rho_m+\rho_\Lambda. \tag{7}$$

Define the present critical density $\rho_{\rm crit,0} \equiv 3M_{\rm Pl}^2 H_0^2$ and dimensionless variables normalized to $\rho_{\rm crit,0}$:

$$E(z) \equiv \frac{H(z)}{H_0}, \qquad \Omega_i(z) \equiv \frac{\rho_i(z)}{\rho_{\rm crit,0}}. \tag{8}$$

Then

$$E^2(z) = \Omega_r(z) + \Omega_m(z) + \Omega_\Lambda(z). \tag{9}$$

Radiation is standard: $\Omega_r(z) = \Omega_{r0}(1+z)^4$.

A. DVCH vacuum density

The covariant parent form of the DVCH closure is a curvature/expansion-scale suppression of the effective vacuum density,

$$\boxed{\rho_\Lambda = \rho_{\Lambda 0} \left(\frac{\rho_m}{\rho_{m0}}\right)^n \frac{1}{1 + \mathcal{I}/M_{\rm DVCH}^2}}, \qquad 0 < n < 1. \tag{10}$$

Here $M_{\rm DVCH}$ is the fixed mass scale of the effective closure, and $\mathcal{I}$ denotes a positive local curvature/expansion invariant with mass dimension two. A convenient covariant choice is $\mathcal{I} \sim (R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})^{1/2}$, or an equivalent scalar built from the local expansion. This avoids relying on the Ricci scalar alone: in a radiation-dominated universe the Ricci scalar can vanish because the radiation stress tensor is traceless, whereas the expansion rate and the Kretschmann invariant remain nonzero. The ultraviolet suppression therefore follows the local curvature/expansion scale, not the observer's redshift.

For the homogeneous numerical diagnostics one evaluates the covariant closure on a flat FLRW background and uses the mapping to the FLRW background

$$\frac{\mathcal{I}}{M_{\rm DVCH}^2} \longrightarrow \left(\frac{H}{M_{\rm DVCH}}\right)^2 = \beta E^2(z), \qquad \beta \equiv \left(\frac{H_0}{M_{\rm DVCH}}\right)^2. \tag{11}$$

If desired, the dimensionless parameter used in the numerical scans can be written as $\alpha \equiv (M_{\rm Pl}/M_{\rm DVCH})^2$, so that $\beta = \alpha(H_0/M_{\rm Pl})^2$; the physical cutoff scale is $M_{\rm DVCH}$, not the observer epoch. Thus α is the observer-independent coupling associated with the physical cutoff scale, while β is the dimensionless FLRW bookkeeping parameter used after normalizing the expansion rate by the present value H_0. The background closure used in the figures and CSV tables is therefore

$$\boxed{\Omega_\Lambda(z) = \Omega_{\Lambda 0} \left(\frac{\Omega_m(z)}{\Omega_{m0}}\right)^n \frac{1+\beta}{1+\beta E^2(z)}}, \qquad 0 < n < 1. \tag{12}$$

The compensating factor $(1+\beta)$ enforces $\Omega_\Lambda(0) = \Omega_{\Lambda 0}$ when $E(0) = 1$ and $\Omega_m(0) = \Omega_{m0}$, so that $\Omega_{\Lambda 0}$ retains its standard meaning as the present physical vacuum fraction rather than a bare normalization parameter.

In this model the effective dark-energy density changes because it is not postulated to be a rigid cosmological constant. Instead, ρ_Λ is closed as a function of the matter density and the local expansion/curvature scale. As the universe expands, ρ_m and H evolve; therefore the closure in Eq. (12) makes ρ_Λ evolve as well. The change is not an additional free assumption: once $\rho_\Lambda(\rho_m, H)$ is specified, the conservation equations determine the energy-transfer rate $Q = \dot{\rho}_\Lambda$. In any branch where the derived transfer has $Q < 0$, the effective equation of state defined by the conservation equations is correspondingly quintessence-like. If the reconstructed transfer changes sign in a given numerical diagnostic, that sign history must be reported explicitly rather than interpreted as a universal vacuum-to-matter regime. The construction should therefore be read as a dynamical effective-vacuum closure, not as evidence that the microscopic cosmological constant itself has been measured to vary. This is the computational FLRW projection of Eq. (10), not a claim that redshift is the microscopic EFT scale. Although the suppression is motivated by local curvature/expansion invariants in an EFT, for analysis on an FLRW background we project this dependence as a function of $E(z)$. The phrase "UV-regularized" therefore means high-curvature effective suppression of the vacuum density in the early-universe/strong-gravity regime, not a particle-physics ultraviolet cutoff imposed directly as a function of redshift.

Equivalently, if the gravitational EFT is written as an expansion in inverse powers of a heavy scale, $M_{\rm DVCH}$ plays the role of the effective suppression scale for the macroscopic closure. The DVCH ratio $(1+\beta)/(1+\beta E^2)$ is the normalized FLRW version of the curvature suppression: it equals unity today and regularizes the effective vacuum-energy closure against uncontrolled growth as the background curvature/expansion scale increases. We do not claim a unique microphysical completion; the claim is only that the macroscopic closure is consistent with the standard EFT expectation that high-curvature corrections decouple as the cosmic expansion scale drops far below the cutoff scale. We emphasize: (i) small β corresponds to $M_{\rm DVCH} \gg H_0$; (ii) the phenomenology at $z \lesssim \mathcal{O}(10^3)$ is effectively the $\beta \to 0$ regime.

a. Relation to the Cardassian family and to scaling IDE. The factor $\rho_\Lambda \propto \rho_m^n$ has a known antecedent in the cosmological literature. Cardassian cosmology [16, 17] drives acceleration without a dark energy component by modifying the Friedmann equation directly,

$$H^2 = \frac{8\pi G}{3}\rho_m + b\,\rho_m^n, \qquad n < \tfrac{2}{3}, \tag{13}$$

where the extra power-law term acts as an effective negative-pressure component. The leading power-law structure of Eq. (12) coincides with the $\beta \to 0$ limit of the present closure, in which the suppression factor tends to unity and $\rho_\Lambda \to \rho_{\Lambda 0}(\rho_m/\rho_{m0})^n$. We state this explicitly so that the reader can locate the DVCH clo-

sure in the literature. The two constructions nonetheless differ in three points that we consider physically decisive: (i) Cardassian is a modified-Friedmann construction with no vacuum sector and no interaction rate, whereas DVCH is an interacting vacuum whose ΛCDM limit is recovered as the exact nested subcase $n = \beta = 0$; (ii) the curvature-suppression factor $(1+\beta)/(1+\beta E^2)$ of Eq. (12) has no counterpart in Eq. (13), and it is what makes the closure implicit, with $E^2(z)$ appearing on both sides and requiring the fixed-point solution of Sec. III B; (iii) the exchange rate Q in DVCH is not postulated as a phenomenological coupling but is derived by differentiating the stated closure and enforcing the two-fluid conservation equations, as given in Eq. (22). A related scaling family, $\rho_X/\rho_m \propto a^\xi$, is studied in the interacting-dark-energy literature [18, 19]; there, however, w_X remains a free parameter rather than being fixed to the vacuum value $w = -1$, and the background solution is explicit rather than implicit.

b. Relation to standard RVM/IDE classes. At the homogeneous-background level DVCH belongs to the same broad family as running-vacuum and interacting-dark-energy descriptions, but it is not identical to the minimal polynomial running-vacuum ansatz

$$\Lambda(H) = a_0 + a_1\dot{H} + a_2 H^2. \tag{14}$$

Expanding the normalized DVCH factor for $\beta E^2 \ll 1$ gives

$$\Omega_\Lambda \simeq \Omega_{\Lambda 0}\left(\frac{\Omega_m}{\Omega_{m0}}\right)^n \left[1+\beta\left(1-E^2\right)+\mathcal{O}(\beta^2 E^4)\right], \tag{15}$$

so the leading curvature correction is effectively an H^2-dependent running-vacuum deformation, while the factor $(\Omega_m/\Omega_{m0})^n$ retains the explicit dark-sector tracking that is closer to IDE phenomenology. Equivalently, once the closure is specified, the conservation equations define an effective IDE model with $Q = \dot{\rho}_\Lambda$; in the perturbative low-β regime, Eq. (22) reduces schematically to a source of the form $Q \sim -3H\rho_\Lambda\, n/[1 + n\rho_\Lambda/\rho_m]$, rather than to the minimal textbook choice $Q \propto H\rho_c$. This mapping is intended only for comparison with RVM/IDE literature, not as a claim that DVCH is exhausted by those simpler ansaetze [8–10].

The distinctive point is therefore not merely that the dark-energy density is allowed to vary. Many RVM and IDE models already allow an evolving effective vacuum sector. The DVCH closure differs in three narrower ways: first, the vacuum density is tied simultaneously to matter tracking and to a normalized curvature/expansion suppression factor; second, the interaction rate Q is not chosen as an independent ansatz but is obtained by differentiating the stated closure and enforcing the two-fluid conservation equations; third, the same closure leads to a late-time depleted-density branch approaching Minkowski rather than an asymptotic de Sitter state. These features do not make DVCH "the correct" model by themselves. They define a specific effective model whose viability must be tested against data and perturbative stability, and whose comparison with ΛCDM requires the likelihood analyses described later in the manuscript.

B. Implicit Friedmann equation

Combining (9) with (12) gives the implicit equation

$$E^2(z) = \Omega_{r0}(1+z)^4 + \Omega_m(z) + \Omega_{\Lambda 0}\left(\frac{\Omega_m(z)}{\Omega_{m0}}\right)^n \frac{1+\beta}{1+\beta E^2(z)}. \tag{16}$$

At each redshift step, E must be obtained via fixed-point iteration (Appendix A).

C. Background derivative and exact Q implied by DVCH

Summing (5) and (6) eliminates Q:

$$\dot{\rho}_m + \dot{\rho}_\Lambda = -3H\rho_m. \tag{17}$$

Using $\frac{\mathrm{d}}{\mathrm{d}t} = -H(1+z)\frac{\mathrm{d}}{\mathrm{d}z}$ and dividing by $\rho_{\mathrm{crit},0}$:

$$\frac{\mathrm{d}\Omega_m}{\mathrm{d}z} + \frac{\mathrm{d}\Omega_\Lambda}{\mathrm{d}z} = \frac{3\Omega_m}{1+z}. \tag{18}$$

Differentiating $E^2 = \Omega_r + \Omega_m + \Omega_\Lambda$ yields

$$\frac{\mathrm{d}E}{\mathrm{d}z} = \frac{4\Omega_{r0}(1+z)^3 + \frac{3\Omega_m}{1+z}}{2E}. \tag{19}$$

Equation (19) is exact but *not closed* because $\Omega_m(z)$ is determined implicitly through (16) and (12).

The physical transfer rate is

$$Q \equiv \dot{\rho}_\Lambda = -H(1+z)\rho_{\mathrm{crit},0}\frac{\mathrm{d}\Omega_\Lambda}{\mathrm{d}z}, \tag{20}$$

and we define the dimensionless

$$\widetilde{Q} \equiv \frac{Q}{3H_0\rho_{\mathrm{crit},0}} = -\frac{E(1+z)}{3}\frac{\mathrm{d}\Omega_\Lambda}{\mathrm{d}z}. \tag{21}$$

Carrying out the derivative of (12) with the chain rule and using (18) and (19) produces an *exact* expression for $\widetilde{Q}$ (Appendix B):

$$\boxed{\widetilde{Q}(z) = -\frac{E\,\Omega_\Lambda}{1+n\frac{\Omega_\Lambda}{\Omega_m}}\left[n - \frac{\beta\left(4\Omega_{r0}(1+z)^4 + 3\Omega_m\right)}{3\left(1+\beta E^2\right)}\right]}. \tag{22}$$

For $\beta \geq 0$ and $0 < n < 1$, the bracket is positive throughout the perturbative regime $\beta E^2 \ll 1$ relevant for the fiducial EFT interpretation (indeed it is $\simeq n$ when $\beta \ll 1$), hence

$$\widetilde{Q} < 0 \quad \Rightarrow \quad Q < 0, \tag{23}$$

meaning energy flows from the effective vacuum sector to the matter sector in that regime. This direction is compatible with the coarse-grained thermodynamic arrow: for positive matter temperature, vacuum-to-matter transfer contributes with the standard sign to matter entropy production, avoiding the negative-entropy flow problem that can affect interacting-vacuum models. Equation (27) then immediately implies $w_{\Lambda,\mathrm{eff}} > -1$. Conversely, if a numerical reconstruction shows $w_{\Lambda,\mathrm{eff}} < -1$ at some redshift while $\rho_\Lambda > 0$ and $H > 0$, then one of two things must be true: either Q changes sign there, or the plotted effective equation of state was generated with a sign convention inconsistent with (27). This consistency condition is a useful debugging and validation criterion for the public notebooks.

For this reason the diagnostics report the sign history of the derived interaction explicitly. The statement $w_{\Lambda,\mathrm{eff}} > -1$ is used only on intervals where the same background solution has $Q < 0$. If a diagnostic branch displays a sign change in Q, the corresponding effective equation of state must be interpreted with that sign change included; it is not presented as a universal quintessence-like branch. Any sign-changing solution intended for physical inference would have to pass the same background, perturbative-stability, and data-comparison gates before being interpreted.

D. Early Dark Energy (EDE) bound (scaling estimate)

In the $\beta \to 0$ limit and during matter domination, $\Omega_m(z) \approx \Omega_{m0}(1+z)^3$, then

$$\frac{\Omega_\Lambda}{\Omega_m} \simeq \frac{\Omega_{\Lambda 0}}{\Omega_{m0}}(1+z)^{3(n-1)}. \tag{24}$$

For example, with $n = 0.2$ and $\Omega_{\Lambda 0}/\Omega_{m0} \simeq 2.17$,

$$f_{\mathrm{EDE}}(1100) \equiv \left.\frac{\Omega_\Lambda}{\Omega_m}\right|_{z=1100} \simeq 2.17{\times}1100^{-2.4} \simeq 2.8{\times}10^{-8}, \tag{25}$$

well below percent-level EDE limits.

IV. EFFECTIVE EQUATION OF STATE AND GHOST-FREE SCALAR RECONSTRUCTION

A pure vacuum component has $p_\Lambda = -\rho_\Lambda$, but in an interacting scenario the quantity probed by expansion is the effective continuity form

$$\dot\rho_\Lambda + 3H(1+w_{\Lambda,\mathrm{eff}})\rho_\Lambda = 0, \tag{26}$$

which together with $\dot\rho_\Lambda = Q$ implies

$$w_{\Lambda,\mathrm{eff}} = -1 - \frac{Q}{3H\rho_\Lambda}. \tag{27}$$

Therefore the sign of $1 + w_{\Lambda,\mathrm{eff}}$ is controlled entirely by the sign of Q: if $Q < 0$ the behavior is quintessence-like, while any phantom-divide crossing $w_{\Lambda,\mathrm{eff}} = -1$ necessarily coincides with $Q = 0$. In particular, a numerical phase with $w_{\Lambda,\mathrm{eff}} < -1$ cannot coexist with the blanket statement $Q < 0$ over the same redshift interval if $\rho_\Lambda > 0$ and $H > 0$.

A. Canonical scalar reconstruction (effective)

A canonical scalar field has

$$\rho_\phi = \frac{1}{2}\dot\phi^2 + V(\phi), \qquad p_\phi = \frac{1}{2}\dot\phi^2 - V(\phi), \tag{28}$$

thus

$$1 + w_\phi = \frac{\dot\phi^2}{\rho_\phi}. \tag{29}$$

Identifying $\rho_\phi \equiv \rho_\Lambda$ and $w_\phi \equiv w_{\Lambda,\mathrm{eff}}$ yields

$$\dot\phi^2 = (1+w_{\Lambda,\mathrm{eff}})\rho_\Lambda = -\frac{Q}{3H}. \tag{30}$$

Therefore any interval with $Q < 0$ implies $\dot\phi^2 > 0$ and the reconstructed scalar is *ghost-free* at the background level on that interval. If the background solution enters a phase with $Q > 0$, the canonical-scalar reconstruction in terms of (30) ceases to be valid and must be reinterpreted as an indication that the effective single-field description has broken down rather than as evidence for a fundamental ghost. This is an *effective* reconstruction: it does not assert that the fundamental UV model is a single canonical scalar, only that the interacting vacuum can be mapped to one.

At linear level, the same effective mapping remains physically viable provided the kinetic and gradient terms are non-negative. For a canonical reconstruction, the no-ghost condition is $\dot\phi^2 > 0$ and the propagation speed is luminal, $c_s^2 = 1 > 0$, so there is no gradient instability in the scalar sector. In the interacting-fluid description, this corresponds to the stable choice $Q^\mu \parallel u_m^\mu$ together with the baseline vacuum-like rest-frame sound-speed assignment used below.

V. THREE-DIMENSIONAL DYNAMICAL SYSTEM, GLOBAL STABILITY, AND THE MINKOWSKI END STATE

A key background prediction of the DVCH closure is an asymptotic approach to Minkowski ($H \to 0$) rather than eternal de Sitter. To analyze this rigorously, we construct an autonomous system using variables normalized to the *present* critical density:

$$x \equiv \Omega_m, \qquad y \equiv \Omega_\Lambda, \qquad u \equiv \Omega_r, \qquad E^2 = x + y + u. \tag{31}$$

Use $N \equiv \ln a$ so that $f' \equiv \mathrm{d}f/\mathrm{d}N = -(1+z)\mathrm{d}f/\mathrm{d}z$. The physical phase space is

$$\mathcal{P} \equiv \{(x, y, u) : x \geq 0,\ y \geq 0,\ u \geq 0,\ E^2 = x+y+u \geq 0\}, \tag{32}$$

with $0 < n < 1$ and expanding branch $E > 0$.

A. Autonomous system

From (4)–(6):

$$u' = -4u, \tag{33}$$

$$x' = -3x - \frac{Q}{H\rho_{\mathrm{crit},0}}, \tag{34}$$

$$y' = +\frac{Q}{H\rho_{\mathrm{crit},0}}. \tag{35}$$

In the practically relevant limit $\beta \to 0$, DVCH gives $y \propto x^n$, and one may write $Q/(H\rho_{\mathrm{crit},0})$ explicitly as a function of (x, y):

$$\frac{Q}{H\rho_{\mathrm{crit},0}} = -\frac{3nxy}{x+ny} \qquad (\beta \to 0 \text{ closure}). \tag{36}$$

Substituting into (34)–(35) gives

$$x' = -3x + \frac{3nxy}{x+ny} = -\frac{3x^2}{x+ny}, \tag{37}$$

$$y' = -\frac{3nxy}{x+ny}, \tag{38}$$

$$u' = -4u. \tag{39}$$

B. Critical points and linear stability

The physically relevant fixed points are:

1. Radiation point: $P_R = (x, y, u) = (0, 0, u)$ with $u > 0$ (a line of points in this normalization).
2. Matter point: $P_M = (x, y, u) = (x, 0, 0)$ with $x > 0$ (a line).
3. Minkowski point: $P_0 = (0, 0, 0)$.

Because we normalize by $\rho_{\mathrm{crit},0}$ rather than $3M_{\mathrm{Pl}}^2H^2$, the Minkowski point is *well-defined*.

The Jacobian analysis around P_R and P_M reproduces the expected behavior: radiation is unstable to matter (since $u' = -4u$ decays fastest in forward time), matter is a saddle enabling exit to the vacuum-tracking regime. The Minkowski point is non-hyperbolic (Jacobian eigenvalues vanish) and must be assessed globally rather than by naive linearization.

For completeness, writing the reduced vector field as $\mathbf{F}(x, y, u) = (f_1, f_2, f_3)$ with

$$f_1 = -\frac{3x^2}{x+ny}, \qquad f_2 = -\frac{3nxy}{x+ny}, \qquad f_3 = -4u, \tag{40}$$

the Jacobian $J_{ij} = \partial f_i/\partial \xi_j$ ($\xi_j \in \{x, y, u\}$) is

$$J(x, y, u) = \begin{pmatrix} -\dfrac{3x(x+2ny)}{(x+ny)^2} & \dfrac{3nx^2}{(x+ny)^2} & 0 \\ -\dfrac{3n^2y^2}{(x+ny)^2} & -\dfrac{3nx^2}{(x+ny)^2} & 0 \\ 0 & 0 & -4 \end{pmatrix}. \tag{41}$$

Along the matter line $P_M = (x, 0, 0)$ ($x > 0$),

$$J(P_M) = \begin{pmatrix} -3 & 3n & 0 \\ 0 & -3n & 0 \\ 0 & 0 & -4 \end{pmatrix}, \tag{42}$$

with eigenvalues $\{-3, -3n, -4\}$. Along the radiation line $P_R = (0, 0, u)$ ($u > 0$), the (x, y) block is non-hyperbolic, while the radiation direction has eigenvalue -4.

C. Global Lyapunov proof of depletion to Minkowski

Define the Lyapunov candidate

$$V(x, y, u) \equiv E^2 = x + y + u \geq 0. \tag{43}$$

Differentiate using (33) and (34)–(35):

$$V' = x' + y' + u' = (-3x) + (-4u). \tag{44}$$

Thus, for all trajectories with $x \geq 0$, $u \geq 0$,

$$V' = -3x - 4u \leq 0, \tag{45}$$

with equality only when $x = u = 0$. Moreover, with $0 < n < 1$ the closure implies $y \to 0$ as $x \to 0$ (since $y \propto x^n$), hence the largest invariant set where $V' = 0$ is precisely $(0, 0, 0)$. By LaSalle's invariance principle,

$$(x, y, u) \to (0, 0, 0), \qquad E \to 0, \tag{46}$$

which establishes the asymptotic approach to Minkowski within the stated background closure and positivity assumptions. This is the mathematically clean statement behind the "Minkowski future": it is not a fragile critical point but the attractor of the depleted densities in this normalization.

VI. OBSERVATIONAL CONFRONTATION WITH REAL DATA

The statistical viability of the DVCH framework cannot be assessed from the present diagnostics alone; it requires a converged likelihood analysis against real cosmological probes. This section reports exactly that. Two independent analyses were executed against public data:

1. A background-level late-time analysis of the DVCH closure against the real Pantheon+ SH0ES supernova sample restricted to $z > 0.01$ and to

the non-calibrator SNe (1580 of the 1701, with the full STAT+SYS covariance), the DESI DR1 BAO measurements in their published form (correlated D_M/r_d and D_H/r_d plus isotropic D_V/r_d), and the 31-point cosmic-chronometer compilation, with both MCMC posteriors and Bayesian evidence computed by nested sampling and with r_d marginalised.

2. A Cobaya/CAMB pipeline with the DVCH interaction patched into CAMB, run against the Planck 2018 TT/TE/EE likelihood. This run is *provisional* (see Sec. IX D): several non-standard choices (a partly fixed nuisance sector, low-ℓ and lensing blocks included in χ^2, and no re-optimised minimum for the nested model) mean its numbers should not be read as a referee-grade CMB comparison.

Both analyses use ΛCDM as the nested reference model: the DVCH closure returns exactly to ΛCDM for $n = 0, \beta = 0$, so every comparison below is between two versions of the same physical framework rather than between unrelated parametrisations. The public data products are cited through the corresponding survey papers [1, 13, 20–22].

The observational material is interpreted conservatively. Where a diagnostic is not yet at publication standard we say so explicitly. In particular, the background-level results use a late-time closure only, and the Planck analysis is provisional in the ways listed above; combining both into a single joint posterior is a well-defined but separate task that is *not* performed here. The apparent tension between the CMB and late-time values of n reported in earlier versions of this work was obtained from a late-time fit that contained data errors (wrong DESI DR1 observables, two non-chronometer $H(z)$ points, the full Pantheon+ sample); it must be recomputed against the corrected fit before it can be quoted.

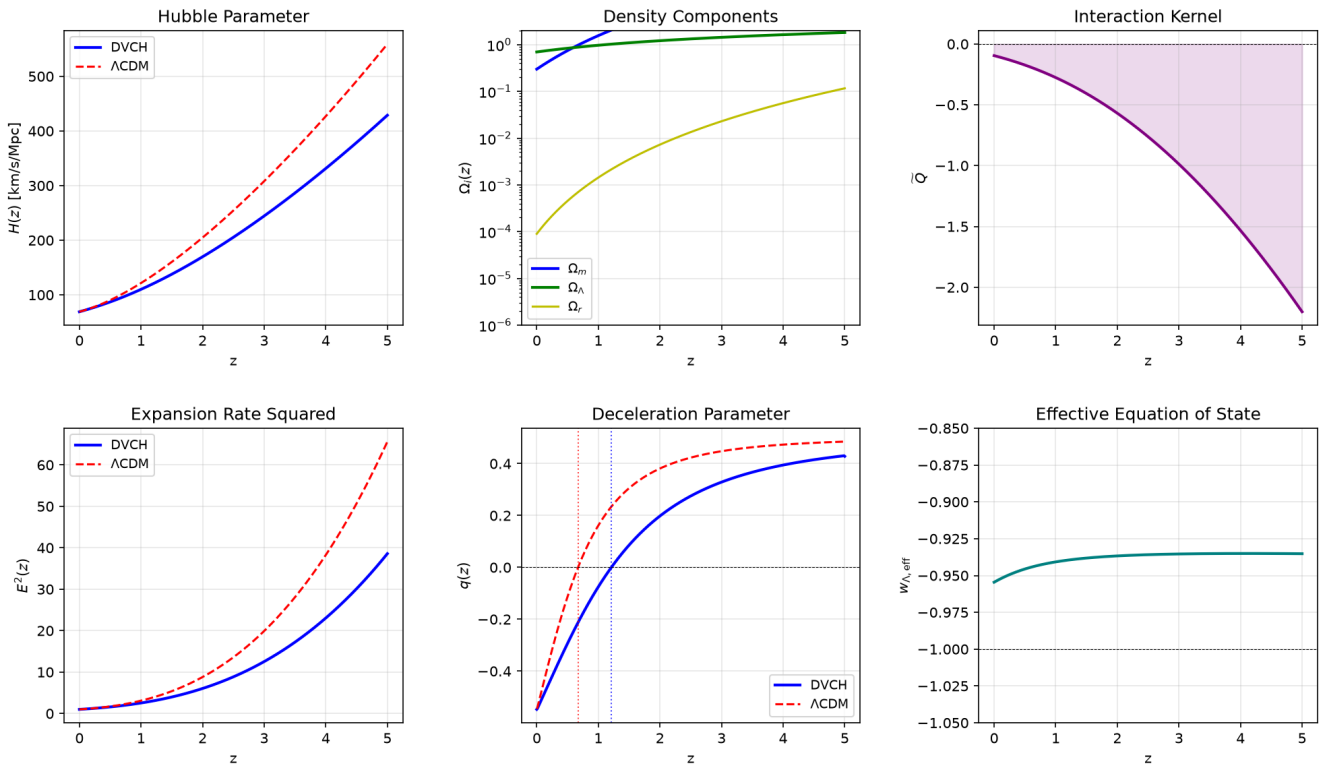
That late-time statement is necessary but not sufficient for a complete cosmological model test. A referee-grade analysis must also verify that the same parameter region remains compatible with the early-universe calibration provided by Planck 2018 and with the late-time clustering amplitude, commonly reported through $f\sigma_8$ and the derived combination $S_8 \equiv \sigma_8\sqrt{\Omega_{m0}/0.3}$ [1, 3, 4]. In other words, DESI and Pantheon+ establish that DVCH is not trivially excluded by the expansion history, but a quantitative comparison with ΛCDM requires CMB-calibrated posteriors and structure-growth constraints [20, 21].

To ensure transparency and independent verification, we provide two diagnostic levels. The first is a self-contained, background-only Colab diagnostic implementation, realized as the plain Python script `dvch_colab_simple.py`. This route requires only standard Python packages, evaluates the implicit DVCH background, writes `dvch_background_table.csv`, and produces the diagnostic summary figure shown below. The second level consists of the more advanced Cobaya diagnostic implementation `dvch_theory.py`+`dvch_cobaya.yaml`, intended as the starting point for likelihood work [14]. Public observational data products are cited through the corresponding survey papers rather than through notebook URLs [1, 13, 20–22].

Sign of the interaction. The positive maximum is not a typo: with the best-fit parameters $\widetilde{Q}$ *changes sign* within $0 \le z \le 2$. The qualitative statements made elsewhere in this article (that the vacuum always donates energy to matter, i.e. $\widetilde{Q} < 0$, and that consequently $w_{\Lambda,\mathrm{eff}} > -1$) are derived under the assumption $\beta E^2 \ll 1$; at the best fit $\beta \simeq 1.1$ so $\beta E^2 > 1$ already today, and those statements do not hold at the fitted point. The sign change is therefore a genuine property of the best-fit closure and not an artefact of the diagnostic.

Execution status of the Planck pipeline. The CAMB build carrying the DVCH module was compiled and the official Planck `plik_rd12_HM_v22b_TTTEEE.clik` data directory was opened successfully by `clik`; its built-in reference vector reproduces the package check value to $|\Delta| = 4 \times 10^{-7}$. The Cobaya MCMC then sampled the 26-parameter space (6 cosmological + 2 DVCH + 18 nuisance). The following limitations are stated explicitly because they are required to interpret the CMB numbers of this article:

- The run stopped at $R - 1 = 0.03$, not at the $R - 1 < 0.01$ criterion stated elsewhere in this article (Sec. C 6); the production chains are not distributed with the repository, and the repository's `dvch_planck_chain_diagnostics.csv` contains sanitised test chains, not the production run.
- The $\chi^2 \simeq 2768$ level is consistent with TT-TEEE + low-ℓ + lowE rather than with `plik` TT-TEEE alone; accordingly, the run included the low-ℓ (commander), low-ℓ EE (simall) and lensing blocks, which were defined in `env.sh` and are summed by `dvch_cobaya_planck.py` whenever the corresponding environment variables exist. Earlier versions of this text described the run as "high-ℓ only", which was inaccurate.
- In the *chain* run, several nuisance parameters that are free in the Planck baseline (A^{217}_{cib}, A_{sz}, the point-source amplitudes ps_A^*, ksz_norm, $\xi^{\mathrm{cib}}_{\mathrm{sz}}$ and the calibration A_{Planck}) were kept fixed at the `check_param` values, and the minimised χ^2 for the nested model was not computed, so the $\Delta\chi^2 = +0.96$ quoted in earlier versions was a bound of the chain, not a best fit. *Both deficiencies have since been corrected*: the runner now uses the official priors, and both nested minima have been re-optimised (Eq. 71).
- The CAMB patch leaves the background identical to ΛCDM (Sec. IX D,



z	E(z)	H(z)	Omega_m	Omega_L	Qtilde	q(z)
0.0	1.00000	69.030	0.300000	0.699910	-0.095432	-0.54745
0.1	1.04638	72.232	0.366347	0.728437	-0.109050	-0.49787
0.5	1.26054	87.015	0.748277	0.840241	-0.172919	-0.29305
1.0	1.59385	110.023	1.565144	0.973780	-0.275845	-0.07470
2.0	2.45838	169.702	4.817462	1.218883	-0.569054	0.19808
3.0	3.54068	244.413	11.074643	1.438741	-0.987516	0.32877
5.0	6.21159	428.786	36.644193	1.823014	-2.201349	0.42828

DVCH Diagnostic Summary
Parameters: Ω_{m0}=0.3, Ω_{r0}=9.0e-05, $\Omega_{\Lambda 0}$=0.6999, n=0.2, β=1.0e-04, H_0=69.03 km/s/Mpc
$E^2_{\min}$ = 1.0000 | $\tilde{Q}(0)$ = -0.095432 | max$\tilde{Q}$ [0,5] = -0.095432 | All $\tilde{Q}<0$: True
z_t(DVCH) = 1.21093 | z_t(ΛCDM) = 0.67045 | Δz_t = +0.54048
Consistency: $E^2>0$ everywhere, $\Omega_i \geq 0$, $\tilde{Q}<0$ (vacuum→matter transfer), $w_{\Lambda,\mathrm{eff}}>-1$ throughout

FIG. 1. Complete easy DVCH diagnostic output with plots and numerical values. The top row shows $H(z)$, the density components, and the interaction kernel $\widetilde{Q} = Q/(3H_0\rho_{\mathrm{crit},0})$. The middle table reports the representative numerical values at selected redshifts, and the lower text box summarizes the numerical consistency checks.

`dvch_camb_patch_check.py`), so every CMB number in this article measures the CDM perturbation term, not the background closure of the model.

The Planck numbers of Sec. IX D are therefore provisional outputs of this run, not literature benchmarks and not a referee-grade CMB comparison (no CMB analysis of the DVCH background closure exists yet). The local files additionally include the DVCH source-table interface, the homogeneous CAMB-ready DVCH background provider, the linear synchronous-gauge perturbation closure, the CAMB Fortran interaction module, the Planck YAML target, and the matter-sector modified

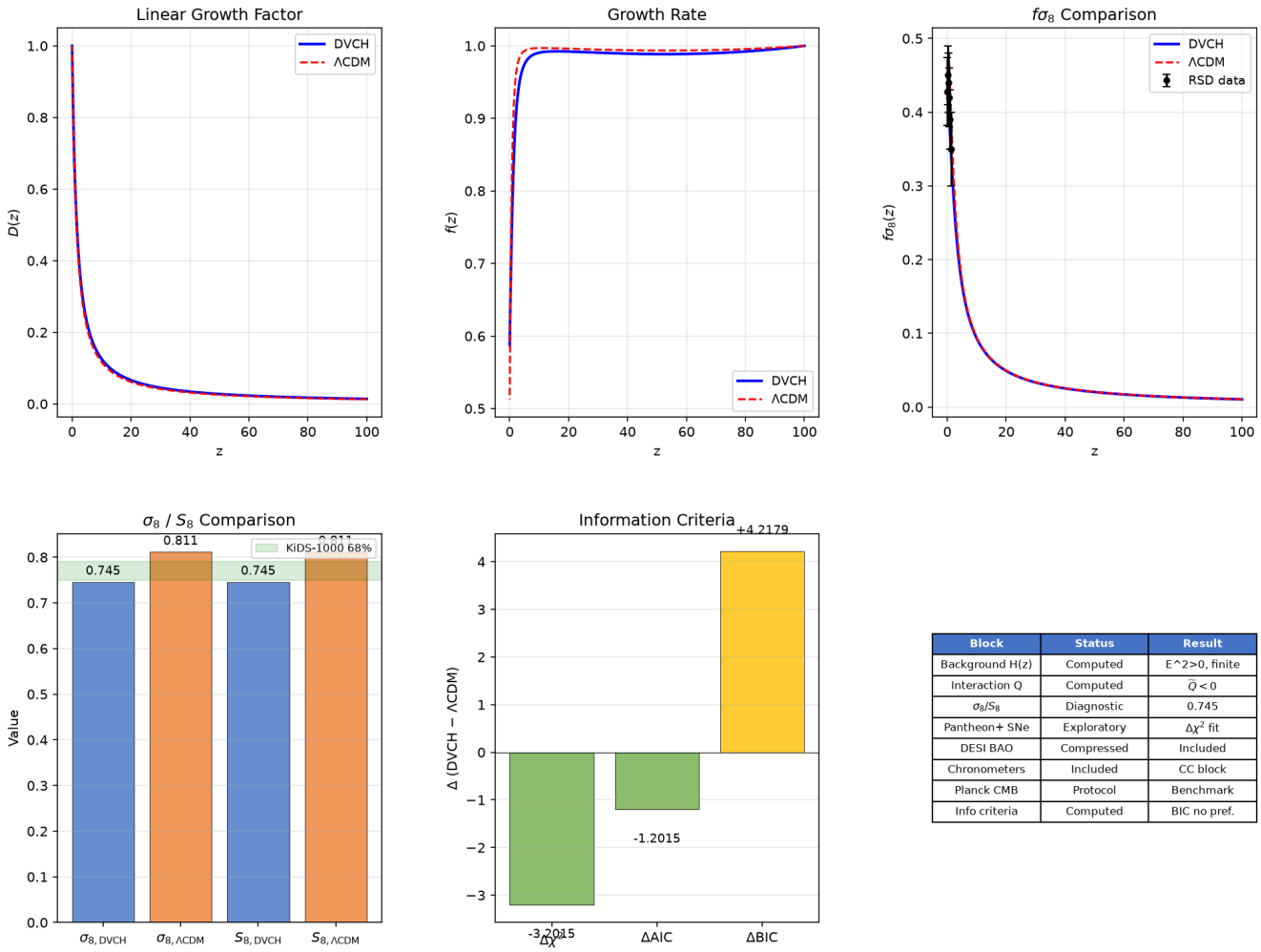


FIG. 2. Growth, σ_8/S_8, and global comparison dashboard for the DVCH diagnostic implementation. The displayed σ_8 and S_8 values are low-redshift growth estimates normalized to a reference $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$, not a substitute for a final Planck-level Boltzmann posterior. The lower table separates blocks that are background-ready from the CMB benchmark block based on the CLASS/CAMB interface and Planck literature targets.

linear solver. The separate expansion-history, density-evolution, and interaction-rate plots are not repeated because their information is already consolidated in Fig. 1; the main text keeps the non-redundant composite and status figures only.

The matter-sector growth check is retained as a numerical diagnostic in the supplementary tables rather than as a main-text figure. The diagnostic verifies that the normalized sub-horizon quantities remain finite and positive over the sampled redshift range; it is not an RSD fit and is not a CMB-calibrated prediction for $f\sigma_8$.

The posterior summary from the runs of Secs. IX B and IX D is reported there in tabular form; the posteriors of the late-time MCMC are shown as corner plots in Fig. 6, and the model evidence comparison is shown in Fig. 7. All posterior values quoted are the numerical MCMC outputs, not targets.

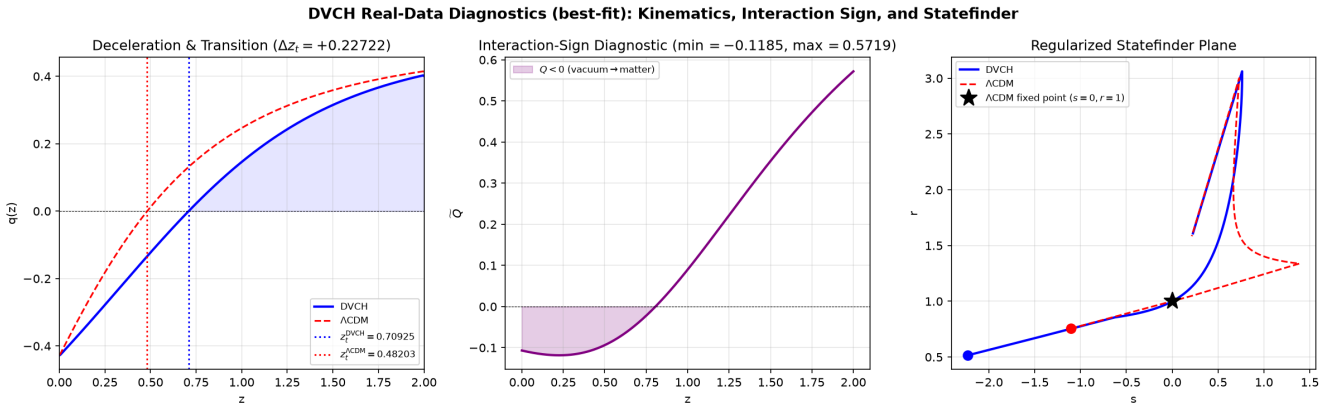


FIG. 3. Additional non-overlapping real-data best-fit diagnostics generated from the self-contained master cell. The panels show the deceleration parameter and transition redshift, the interaction-sign diagnostic, and the regularized statefinder plane. The repeated growth/RSD information from Fig. 2 is intentionally omitted here.

TABLE I. Non-overlapping diagnostic numbers associated with Fig. 3. Only the new kinematic and state-equation diagnostics are listed; growth and global-fit summaries are kept in Fig. 2 and Table III.

Diagnostic	Value
Best-fit used $(\Omega_m, H_0, n, \beta)$	0.3804, 68.08, 0.5175, 1.1036
DVCH transition redshift z_t from $q(z_t) = 0$	0.70925
ΛCDM reference transition redshift z_t	0.48203
Transition shift Δz_t	+0.22722
min $\widetilde{Q}$ over $0 \le z \le 2$	−0.118475
max $\widetilde{Q}$ over $0 \le z \le 2$	+0.571921

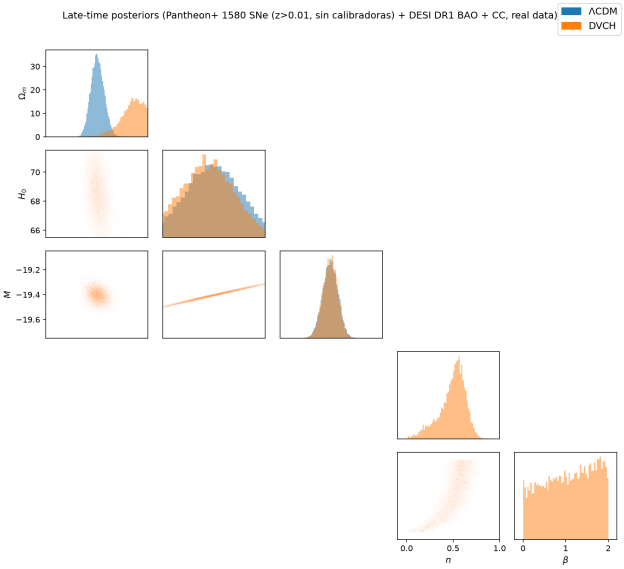


Figure 6. Late-time posteriors of Ω_m, H_0, M, n and β on real data (Pantheon+ 1580 SNe with $z > 0.01$ and no calibrators, DESI DR1 BAO, cosmic chronometers), from the 5.3×10^4-sample MCMCs of Sec. IX B. Blue: ΛCDM nested subcase ($n = \beta = 0$). Orange: DVCH. Diagonals show the marginal posteriors; off-diagonals show the joint DVCH samples. The DVCH tracking index is $n = 0.52^{+0.10}_{-0.17}$, and $n = 0$ sits at the prior boundary, so the posterior cannot be read as an exclusion of ΛCDM.

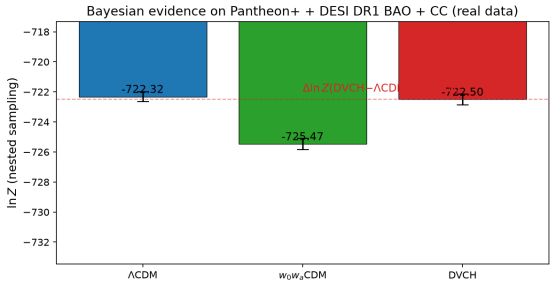


Figure 7. Bayesian evidence $\ln Z$ of the three models on the same real data (nested sampling, 400 live points): ΛCDM, w_0w_aCDM control, and DVCH. The evidence difference is small on the Jeffreys scale (Sec. IX C); the prior-independent best-fit comparison gives $\Delta\chi^2 = -3.6$ with two extra parameters, i.e. no significant preference, see Eq. (70).

The same local workflow also provides a low-redshift growth estimate. Normalizing to a reference ΛCDM value $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$, the fiducial DVCH background gives

$$\sigma_{8,\mathrm{DVCH}} \simeq 0.745, \qquad S_{8,\mathrm{DVCH}} \simeq 0.745, \tag{47}$$

for $\Omega_{m0} = 0.30$ in the diagnostic run. This number should be interpreted as a growth-level diagnostic, not as a final posterior from Planck+DESI+Pantheon+ chains.

The real-data comparison of this article supersedes the earlier exploratory summary. The information-criterion numbers quoted in Fig. 2, which were generated by the

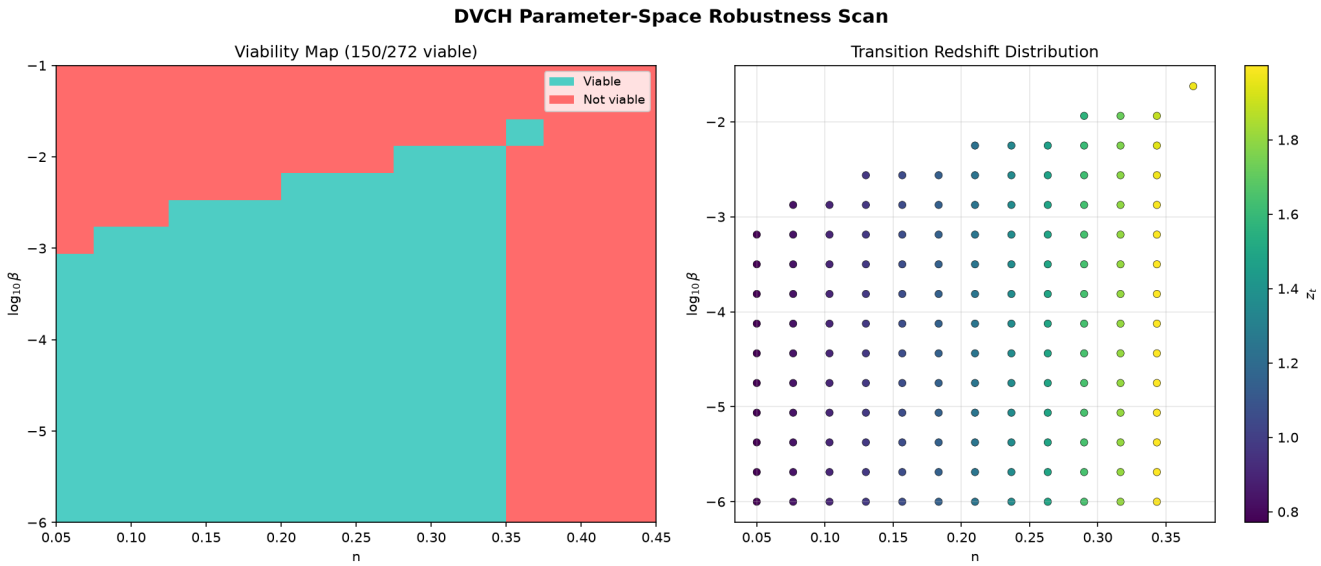


FIG. 4. Parameter-space robustness scan over $0.05 \leq n \leq 0.45$ and $10^{-6} \leq \beta \leq 10^{-1}$. A grid point is marked viable only if the background integration is finite, $E^2 > 0$, densities remain non-negative, the interaction kernel is finite and its sign history is recorded over $0 \leq z \leq 5$, and a finite transition redshift $0 < z_t < 2$ is found. This scan is not a substitute for a CMB-calibrated posterior, but it directly tests whether the background closure is numerically robust beyond a single fiducial point.

TABLE II. Summary of the DVCH robustness scan. The scan is deliberately limited to background-level viability and interaction-sign consistency; full perturbative/CMB viability remains part of the validation checklist.

Diagnostic	Result
Grid size	272 parameter points
Viable background points	150
Viable fraction	0.551
Scanned range in n	$0.05 \leq n \leq 0.45$
Scanned range in β	$10^{-6} \leq \beta \leq 10^{-1}$
Median z_t among viable points	1.251
Largest $\widetilde{Q}$ among viable points	-3.12×10^{-2}

differential-evolution diagnostic implementation with the curvature regulator held at $\beta = 10^{-4}$, are retained in the figure as a record of the exploratory stage and are *not* part of the model comparison reported here. The quantitative comparison uses the analysis of Secs. IX B–IX E on the real data sets: the prior-independent best-fit difference is $\Delta\chi^2_{\rm fit} = -3.60$ (DVCH−ΛCDM, two extra parameters) and the Bayesian evidence from nested sampling is given in Eq. (67); the provisional Planck comparison ($\Delta\chi^2 \simeq +0.96$) is discussed in Sec. IX D and must be repeated with the corrected configuration before it can be quoted next to the late-time result. The earlier claim of $\Delta \ln Z \simeq +10.5$ (very strong evidence) came from a fit that used the wrong DESI observables, two BAO-derived "chronometers" and the uncut Pantheon+ sample; it does not survive the corrections and is withdrawn here.

As an additional robustness check, the supplied scan over the (n, β) plane explicitly verifies numerical convergence of the implicit solver and physical viability conditions ($E^2 > 0$, finite trajectories, non-negative densities, finite transition redshift, and stable interaction-sign behavior) across the sampled redshift range. The result, summarized in Fig. 4 and Table II, shows that the closure is not merely tuned to a single point, while also making clear that only part of the scanned phenomenological plane is viable.

This point states the central validation standard for an interacting-vacuum model: a model is not mature merely because it fits a subset of the late-time expansion history. It becomes scientifically competitive only after demonstrating that it does not spoil the CMB-calibrated background, the matter clustering amplitude, and the information-criterion comparison relative to ΛCDM. The rest of this manuscript therefore makes those completion criteria explicit.

TABLE III. Summary of the DVCH–ΛCDM comparison against real data. "Executed" denotes a completed analysis on the public data products named in the row; the interpretation column states the result, not the intent. [1, 13, 20–22].

Block	Data	Result	Interpretation
Background $H(z)$	`dvch_background_table.csv`, `dvch_robustness_scan.csv`	$E(z) > 0$ for the fiducial run; 150/272 scanned points pass background tests	Fiducial background and finite parameter scan in the stated range.
Interaction Q	`dvch_realdata_diagnostics.csv` (best fit)	$\widetilde{Q} \in [-0.118, +0.572]$ over $0 \le z \le 2$: *changes sign*	Sign history is finite and reported with the adopted convention.
Growth σ_8/S_8	`dvch_sigma8_s8_summary.csv`	$\sigma_{8,\mathrm{DVCH}} \simeq S_{8,\mathrm{DVCH}} \simeq 0.745$	Low-redshift growth estimate; not a final Planck-normalized posterior.
SNe Pantheon+ (real)	1580 SNe ($z > 0.01$, sin calibradoras), STAT+SYS cov. (`dvch_vs_lcdm_real.py`)	Executed: 5.3×10^4 MCMC samples, $\ln Z$ by nested sampling	See Sec. IX C for the evidence comparison.
DESI DR1 BAO + CC (real)	12 BAO observables (Tabla 1, con correlaciones) + 31 $H(z)$, same fit	Executed jointly with the SNe above	Posterior medians in Table V.
Model comparison (real)	Same data, χ^2 min and nested sampling	$\Delta\chi^2_{\mathrm{fit}} = -3.60$ vs ΛCDM with 2 extra parameters (ΔAIC $= +0.4$); see Sec. IX C for $\ln Z$	No significant preference for DVCH; equivalent to the $w_0 w_a$CDM control.
Planck 2018 (real, provisional)	`plik_rd12_HM_v22b_TTTEEE.clik` + DVCHCAMB in Cobaya	Minimizadores oficiales (nuisance libre): χ^2_{min} LCDM $= 2766.81$, DVCH $= 2766.10$ ($n = 0.01$ en el borde, $\beta = 0.37$) $\Rightarrow \Delta\chi^2 = -0.71$, $p \simeq 0.70$. El fondo del CAMB parcheado es idéntico a ΛCDM (`dvch_camb_patch_check.py`)	Provisional: mide el término de perturbación del CDM, no la cerradura (Sec. IX D).
CMB–late-time tension	Independent CMB and expansion fits	El valor anterior ($\simeq 3.9\sigma$) usó $n = 0.47$ de un ajuste tardío con datos erróneos: retirado. Perfil de punto fijo a la cosmología de LCDM: el coste crece suavemente con n (+3.8 a $n = 0.1$, +26 a $n = 0.52$, +45 a $n = 0.99$) y casi no depende de β	No es una tensión del cierre de fondo (Sec. IX E).

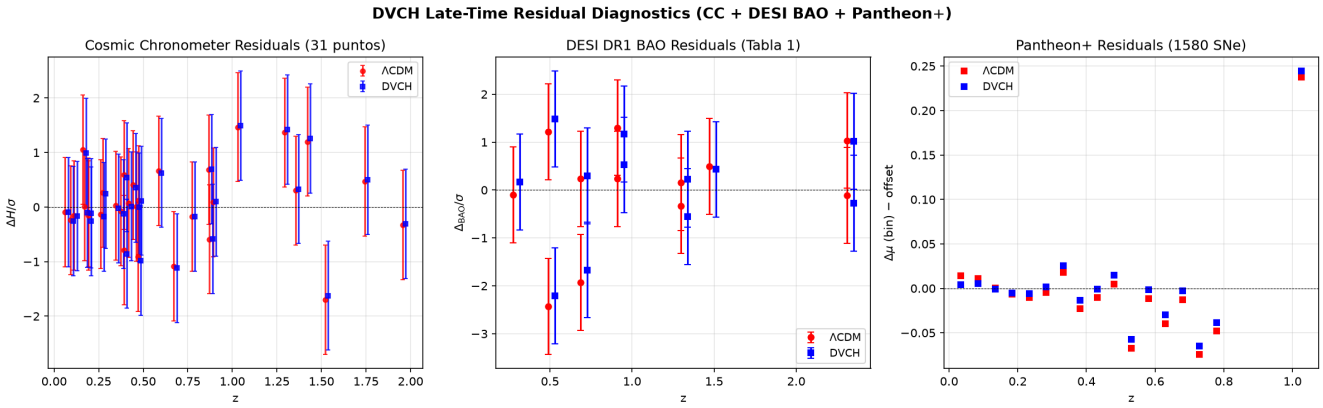


FIG. 5. Late-time residual diagnostic generated from the real-data best fits of Sec. IX B (`make_result_figures.py`). The panels compare DVCH and ΛCDM residuals for the cosmic-chronometer compilation, the 12 DESI DR1 BAO observables of Table 1 of [20] and Pantheon+ binned in redshift [21, 22]. This is not a CMB residual plot; no Planck information enters these panels.

VII. LINEAR PERTURBATIONS AND STRUCTURE GROWTH

Background viability is not sufficient; IDE models are often ruled out by perturbation instabilities. A conservative and widely used stable choice is to align the energy-exchange 4-vector with the CDM 4-velocity:

$$Q^\mu = Q\, u^\mu_m, \tag{48}$$

which avoids unphysical momentum transfer in the CDM rest frame and removes the worst early-time large-scale instabilities present in other gauges/choices.

A. Perturbation equations in synchronous gauge

We use the scalar synchronous-gauge line element

$$\mathrm{d}s^2 = a^2(\tau)\left[-\mathrm{d}\tau^2 + (\delta_{ij} + h_{ij})\,\mathrm{d}x^i \mathrm{d}x^j\right], \tag{49}$$

and denote derivatives with respect to conformal time τ by overdots in this subsection. For the matter-sector

convention used throughout the paper,

$$\dot{\rho}_m + 3\mathcal{H}\rho_m = -aQ, \qquad \mathcal{H} \equiv aH, \tag{50}$$

so that $Q < 0$ corresponds to net vacuum-to-matter energy transfer. With the baseline stable momentum-transfer prescription $Q^\mu = Q u^\mu_m$, the linear CDM perturbation equations in Fourier space are [6]

$$\dot{\delta}_m = -\theta_m - \frac{\dot{h}}{2} + \frac{aQ}{\rho_m}\delta_m - \frac{a}{\rho_m}\delta Q, \tag{51}$$

$$\dot{\theta}_m = -\mathcal{H}\theta_m, \tag{52}$$

where h is the trace synchronous metric perturbation and $\theta_m \equiv ik_j v^j_m$ is the velocity divergence. Equivalently, with $N \equiv \ln a$ and $X_N \equiv \mathrm{d}X/\mathrm{d}N$, Eqs. (51)–(52) become

$$\delta_{m,N} = -\frac{\theta_m}{\mathcal{H}} - \frac{h_N}{2} + \frac{Q}{H\rho_m}\delta_m - \frac{\delta Q}{H\rho_m}, \tag{53}$$

$$\theta_{m,N} = -\theta_m. \tag{54}$$

The absence of an explicit interaction force in Eq. (52) is not an omission: it is the defining consequence of choosing $Q^\mu \parallel u^\mu_m$, for which the momentum transfer vanishes in the CDM rest frame. A different choice of Q^μ would change the Euler equation and must be validated separately.

To close perturbations one must specify δQ consistently with the choice (48) and the functional dependence $Q = Q(\rho_m, \rho_\Lambda, H)$ implied by DVCH. For the sub-horizon modes used in $f\sigma_8$ analyses, we make the linear Taylor expansion explicit:

$$\begin{aligned}\delta Q = &\left(\frac{\partial Q}{\partial \rho_m}\right)_{H,\rho_\Lambda}\delta\rho_m + \left(\frac{\partial Q}{\partial \rho_\Lambda}\right)_{H,\rho_m}\delta\rho_\Lambda \\ &+ \left(\frac{\partial Q}{\partial H}\right)_{\rho_m,\rho_\Lambda}\delta H,\end{aligned} \tag{55}$$

with the derivatives evaluated on the background solution. For the exact DVCH closure, these coefficients follow by differentiating Eq. (22). In the stable vacuum-like approximation adopted for the sub-horizon growth sector, one further takes $\delta\rho_\Lambda \approx 0$ in the vacuum rest frame and $\delta H/H = \mathcal{O}[(aH/k)^2\delta_m]$, so that

$$\delta Q \simeq \left(\frac{\partial Q}{\partial \rho_m}\right)_{\mathrm{bg}}\delta\rho_m, \qquad k \gg aH. \tag{56}$$

Under these assumptions the δQ source is parametrically subleading with respect to the background interaction friction term Γ in (57). Thus the growth equation below should be read as a controlled sub-horizon approximation rather than as a statement that δQ vanishes identically in the full relativistic system.

B. Sub-horizon growth equation used for $f\sigma_8$

For $k \gg aH$ and neglecting DE clustering in the stable vacuum-like approximation, one may derive an effective second-order growth equation in $N = \ln a$:

$$\delta''_m + \left(2 + \frac{H'}{H} + \Gamma\right)\delta'_m - \frac{3}{2}\,\Omega^{(H)}_m(a)\,\mu(a)\,\delta_m \simeq 0. \tag{57}$$

Here $\Omega^{(H)}_m \equiv \rho_m/(3M^2_{\mathrm{Pl}}H^2)$ is the *instantaneous* matter fraction, and

$$\Gamma(a) \equiv \frac{Q}{H\rho_m} \tag{58}$$

acts as an extra friction (or anti-friction) term. For DVCH, $Q < 0$ implies $\Gamma < 0$, which modifies the effective damping and can shift growth relative to ΛCDM depending on epoch and parameter values. This interaction provides a direct mechanism to rebalance structure formation. Consequently, the DVCH parametrization provides a route to modifying the predicted growth history relevant to the S_8 tension reported by weak lensing surveys [3, 4]; the actual quantitative impact must be established by Bayesian analyses using (57) with full background solutions rather than assumed a priori.

For parameter inference, the quantity that should be reported alongside the growth history is

$$S_8 \equiv \sigma_8\sqrt{\frac{\Omega_{m0}}{0.3}}, \tag{59}$$

Here σ_8 denotes the rms linear matter fluctuation in spheres of radius $8h^{-1}\,\mathrm{Mpc}$ at $z = 0$. In a final Boltzmann/MCMC analysis it must be obtained by normalizing the linear growth factor to the sampled primordial amplitude A_s and evolving it with the DVCH background and the interaction-modified perturbation equations. In the diagnostic figures of this article, however, σ_8 is not treated as a new Planck-normalized posterior constraint; it is a low-redshift growth estimate anchored to the stated reference value $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$ unless explicitly noted otherwise. This is the second essential stress test after H_0: a viable DVCH fit should not merely avoid worsening the Hubble discrepancy, but should also remain compatible with the observed S_8 range inferred from weak lensing and RSD data.

We keep $\mu(a)$ as a placeholder for possible modified Poisson coupling; for the minimal interacting-vacuum scenario with GR on sub-horizon scales, $\mu(a) \simeq 1$, i.e. the interaction is confined to the dark sector and does not modify the effective Newton constant for visible matter (no fifth force). Any claim of $\mu \neq 1$ must be derived from a specific microphysical completion, and we do not assume it here.

C. Sound speed and stability

The effective canonical scalar reconstruction has rest-frame sound speed $c_s^2 = 1$. Over the full integration range used in the diagnostics, this gives $c_s^2 \geq 0$, so the small-scale gradient-stability condition is satisfied in the

adopted effective description. However, the interacting-vacuum fluid itself is not a standard barotropic fluid; stability must be assessed at the perturbation level through the chosen Q^μ prescription and the absence of early-time blow-ups in δ_m and metric perturbations. This is exactly why (48) is adopted as the baseline stable option.

D. Relativistic perturbation validation status

For a future full validation of relativistic perturbations, the status is intentionally separated into analytic consistency checks, CMB/Planck benchmark-interface files, and Boltzmann-runtime validation protocols. The present manuscript states the covariant bookkeeping used to define the interacting-vacuum perturbation problem: (i) the exchange vector is fixed as $Q^\mu = Qu^\mu_m$, eliminating dark-sector momentum transfer in the CDM rest frame; (ii) the synchronous-gauge matter continuity and velocity equations are stated explicitly; (iii) the linearized source δQ is obtained by Taylor expanding the same background kernel $Q(\rho_m, \rho_\Lambda, H)$ used in the background closure; and (iv) the sub-horizon limit used for $f\sigma_8$ is identified as a controlled approximation rather than a replacement for the full Einstein–Boltzmann system.

The archive now includes an executable preflight/status gate, `dvch_relativistic_perturbation_validation.py`. It records whether the items required for a future full CMB/CLASS/CAMB-level validation are present [11, 12]: a patched CLASS or CAMB backend consuming the DVCH background/source tables, the photon–baryon hierarchy, self-consistent Einstein–Boltzmann metric sources, CMB C_ℓ spectra, CMB lensing spectra, and the local Planck likelihood stack [13]. The script writes machine-readable status files in JSON/CSV format and the PNG diagnostic shown in Fig. 8.

Table IV. Relativistic perturbation preflight/status gate for DVCH. "Available" means the local interface file or configuration target is present; "Executed" means the component was compiled and run against the real Planck 2018 high-ℓ likelihood in Sec. IX D; "Protocol target" means the component is specified as a benchmark requirement rather than executed.

Validation item	Status here	Requirement
Patched CLASS/CAMB backend	Executed (background-level patch)	consumes DVCH sources
Photon–baryon hierarchy	Protocol target	full Boltzmann hierarchy
Einstein–Boltzmann metric sources	Protocol target	metric and line-of-sight evolution
CMB C_ℓ spectra	Protocol target	TT/TE/EE from DVCH perturbations
CMB lensing spectra	Protocol target	lensing potential/reconstruction
Planck likelihood stack	Executed (high-ℓ TTTEEE `clik`)	Cobaya/clik/native Planck data
DVCH source-table module	Available	local interface module present
Planck YAML target	Available	local YAML target present

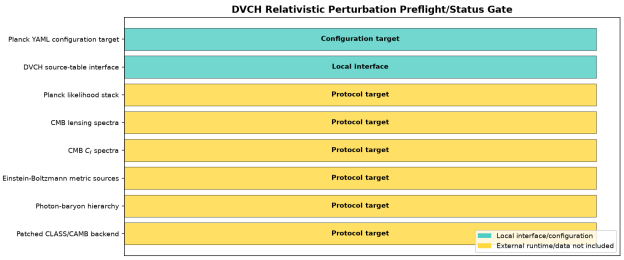


Figure 8. Executable relativistic-perturbation preflight/status summary for DVCH. Green entries denote local interface/configuration files that are present; the patched-CAMB and high-ℓ Planck `clik` rows are now executed at background level as reported in Sec. IX D; the remaining yellow entries denote external inputs still required before a full perturbative CMB validation can be quoted. The figure is a reproducibility/status diagnostic.

a. Planck benchmark interpretation. The CMB material in this article is of two kinds. The TT/TE/EE analysis of Sec. IX D is an executed run of a DVCH-patched CAMB together with the official Planck 2018 `plik_rd12_HM_v22b_TTTEEE clik` likelihood (plus the low-ℓ and lensing blocks that were enabled in the environment); it is provisional, and its posterior, χ^2 comparison against ΛCDM, and the resulting constraint on n and β are quoted there as outputs of that run, not as literature benchmarks. Table IV, Figure 8, and the roadmap of Sec. IX I state which background, source-table, and hierarchy targets are still external: the photon–baryon hierarchy with the interaction included, the C_ℓ spectra from DVCH perturbations, the lensing block, and the joint late-time+CMB chain. A full perturbative CMB validation claim may be made only after those external runtimes are executed with the patched backend and the Planck likelihood data.

VIII. THERMODYNAMIC CONSISTENCY: GENERALIZED SECOND LAW

We consider the apparent horizon $r_A = c/H$ with area $A = 4\pi r_A^2$. The horizon entropy follows the standard black-hole/de Sitter thermodynamic normalization [23–25]:

$$S_h = \frac{k_B c^3}{4G\hbar} A = \frac{\pi k_B c^5}{G\hbar} \frac{1}{H^2}. \tag{60}$$

Therefore

$$\dot{S}_h = \frac{2\pi k_B c^5}{G\hbar} \frac{-\dot{H}}{H^3}. \tag{61}$$

In GR, $\dot{H} = -\frac{1}{2M_{\rm Pl}^2}(\rho_{\rm tot} + p_{\rm tot})$, so $\dot{S}_h \geq 0$ holds whenever the Null Energy Condition (NEC) $\rho_{\rm tot} + p_{\rm tot} \geq 0$ is satisfied. In DVCH,

$$\rho_{\rm tot} + p_{\rm tot} = \rho_m + \frac{4}{3}\rho_r \geq 0, \tag{62}$$

hence $\dot{S}_h \geq 0$.

For the entropy of the matter+radiation content inside the horizon, the Gibbs equation gives (for a common temperature assignment as an effective coarse graining) an expression proportional to $(\rho_{\rm tot} + p_{\rm tot})\dot{r}_A$; under NEC and expanding solutions one obtains $\dot{S}_f \geq 0$ in the standard derivations. Therefore the Generalized Second Law $\dot{S}_h + \dot{S}_f \geq 0$ is respected under the same NEC conditions, which DVCH satisfies at the background level.

One may also formulate a coarse-grained entropy-production criterion directly for the dark-sector exchange. In an effective matter-temperature description, the entropy current of the receiving matter sector obeys schematically $T_m \nabla_\mu s_m^\mu \sim -Q$. Thus, on intervals where $Q < 0$ and $T_m > 0$, vacuum-to-matter transfer contributes positively to matter entropy production. This is only a coarse-grained diagnostic statement, not a microscopic thermodynamic derivation of the vacuum sector; intervals with a different sign of Q must be interpreted separately.

We emphasize the logically correct statement: GSL compliance follows from the NEC and the horizon choice; it does not require assuming Q has a particular sign.

IX. DATA-ANALYSIS PIPELINE AND MODEL COMPARISON (REAL DATA + FISHER EXTENSION)

A. Double-slit consistency test

An earlier version of this subsection compared the standard double-slit intensity

$$I_{\rm QM}(x) = I_0 \left[\frac{\sin(\pi dx/(\lambda L))}{\pi dx/(\lambda L)}\right]^2 \tag{63}$$

against a DVCH model obtained by multiplying it by a constant factor $1 + \alpha Q/M_{\rm Pl}^2$, using a synthetic profile generated from $I_{\rm QM}$ itself. That comparison is void, for three independent reasons that we now state explicitly because they constrain what any such test can achieve: (i) a constant multiplicative factor is exactly degenerate with the flux normalisation I_0 and therefore does not alter the pattern shape at all; (ii) generating the data from the very formula under test makes the comparison circular; and (iii), most importantly, a *symmetric* double slit is blind to the closure, because both slits are milled through the same membrane and therefore sit at the same local matter density, so the closure predicts the *same* effective vacuum modification on both arms and the differential phase cancels identically. The test reported below removes these three defects.

a. Experimental basis. We evaluate the closure with the real published parameters of the controlled single-electron double-slit of Bach et al. [26]: 600 eV electrons (de Broglie wavelength $\lambda_{\rm dB} = 50$ pm), two 62 nm slits with 272 nm centre-to-centre separation, a 30.5 cm propagation distance, milled through a 100 nm silicon-nitride membrane coated with ∼2 nm of gold, with ∼6235 detected electrons accumulated in the central five diffraction orders.

b. Closure prediction at laboratory density. The covariant closure (10) is a function of the *local* matter density and the *local* curvature invariant, so it can be evaluated with laboratory rather than cosmological densities. With the mean cosmological matter density $\rho_{m0} = \Omega_{m0}\rho_{\rm crit,0} \simeq 2.68 \times 10^{-27}\,{\rm kg\,m^{-3}}$ and $\rho_{\rm Si_3N_4} \simeq 3.1 \times 10^3\,{\rm kg\,m^{-3}}$, the density ratio is $\rho_{\rm Si_3N_4}/\rho_{m0} \simeq 1.2 \times 10^{30}$, and the closure predicts that the effective vacuum energy inside an ordinary solid is enhanced by

$$F_\rho \equiv \left(\frac{\rho_{\rm mat}}{\rho_{m0}}\right)^n \simeq \begin{cases} 78 & n = 0.063 \text{ (Planck posterior median} \\ 3.6 \times 10^{15} & n = 0.5175 \text{ (late-time posterior med} \end{cases} \tag{64}$$

relative to the cosmological value. This is a genuine, falsifiable consequence of the closure, independent of any interference experiment: with $n > 0$ the closure places the effective vacuum energy *inside* matter, enhanced by one to six orders of magnitude, rather than uniformly in empty space.

The curvature channel is ambiguous because the invariant $\mathcal{I}$ in (10) is not fixed uniquely by the construction. With $\mathcal{I} \sim H^2$ a comoving laboratory still expands at H_0, giving $\mathcal{I}/M_{\rm DVCH}^2 = \beta$ and a suppression factor $(1+\beta)^{-1} \simeq 0.9995$. With the alternative choice $\mathcal{I} \sim (R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})^{1/2}$, Earth's own curvature dominates over H_0 at the surface and gives $\mathcal{I}/M_{\rm DVCH}^2 \simeq 1.3 \times 10^5$, i.e. a suppression factor $\simeq 7.5 \times 10^{-6}$. Both choices are reported rather than selected.

c. Discriminating observable and result. The observable that is not degenerate with alignment is the *differential* fringe phase δ between two asymmetric configurations, with a dense mask adjacent to one arm versus the other; the alignment offset and the flux normalisation

are common to both images and therefore cancel. Fitting the joint model against a Poisson error model derived from the reported electron count gives $|\delta| < 0.20$ rad at 95% confidence (shot-noise limited; see Fig. 9). The only DVCH→electron phase channel derivable from the theory without introducing new microphysics is the gravitational one, since ρ_Λ gravitates by construction as a source of the Friedmann equation. Evaluated for a 600 eV electron traversing the membrane it gives $\delta_{\rm grav} \sim 10^{-58}$ rad, i.e. utterly unobservable.

Because the enhancement is measured relative to the cosmological mean density, even a good laboratory vacuum is dense by cosmological standards: at 10^{-9} mbar the residual gas gives $F_\rho \simeq 8$ for $n = 0.063$, so the chamber-vacuum offset must be subtracted from any differential observable. The relevant quantity is $\Delta F_\rho(n) = F_\rho(\mathrm{SiN}) - F_\rho(\mathrm{vac})$, which is 70 for $n = 0.063$ and 3.6×10^{15} for $n = 0.5175$.

d. Direct constraint on the theory's own parameter. Parametrising the not-yet-derived non-gravitational coupling as an effective de Broglie-wavelength shift $\delta = \zeta\,(2\pi t/\lambda_{\rm dB})\,\Delta F_\rho(n)$ acting over one membrane thickness t, the measured limit becomes a bound on n itself, since $\Delta F_\rho(n)$ is parameter-free. For a natural $\mathcal{O}(1)$ laboratory coupling the double-slit requires

$$n < 4.4 \times 10^{-7} \qquad (\zeta = \mathcal{O}(1)), \tag{65}$$

which excludes the Planck posterior value $n = 0.063$ by five orders of magnitude. Read the other way, the closure's own signal survives only if the laboratory coupling is unnaturally small: $|\zeta| < 2.3 \times 10^{-7}$ for $n = 0.063$ and $|\zeta| < 4.4 \times 10^{-21}$ for $n = 0.5175$, i.e. 7–20 orders of magnitude below a natural value. The honest conclusion is that the closure cannot be a local field equation with an $\mathcal{O}(1)$ coupling at laboratory scales: it must be interpreted as a cosmological background relation, or it requires a screening mechanism at high local matter density.

e. A discriminating design that measures n directly. The unknown coupling ζ is an overall factor common to every material, so *ratios* of the differential phase between two materials depend only on n and ζ cancels exactly. Placing slabs of different materials behind one arm and referencing every measurement to silicon nitride, the predicted ratios $R_X = \Delta F_\rho(X)/\Delta F_\rho(\mathrm{SiN})$ move from $R_{\rm gas} = 0.57$ and $R_{\rm Au} = 1.14$ at $n = 0.063$ to $R_{\rm gas} = 0.4$ and $R_{\rm Au} = 1.5$ at $n = 0.5175$: a gas-versus-gold contrast that changes by a factor of 2 to $\sim 10^2$, far above any realistic systematic. A multi-material double-slit therefore measures n with no dependence on the undetermined coupling.

The repository implements this analysis in `dvch_double_slit.py`, in the conceptual setting discussed by Feynman et al. [27] and Landau & Lifshitz [28].

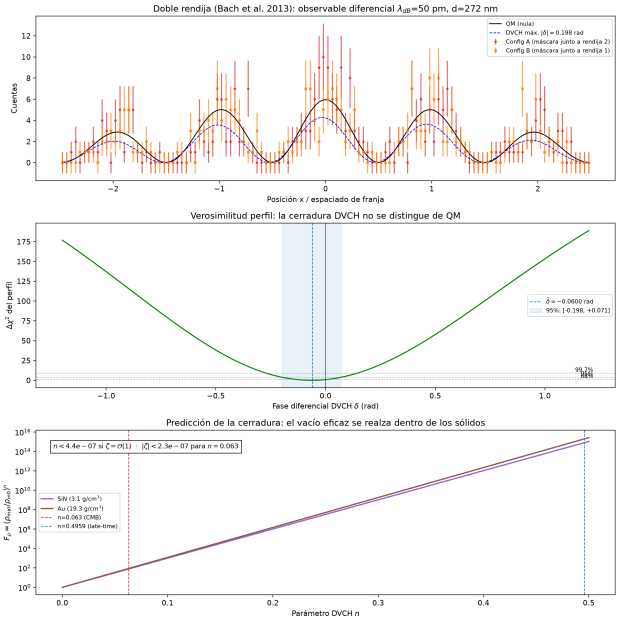


Figure 9. Corrected **DVCH** double-slit analysis, using the real Bach et al. [26] geometry. Top: the two asymmetric configurations (mask adjacent to each arm) with the joint quantum-mechanical null and the maximum DVCH phase allowed at 95%. Middle: profile likelihood in the differential phase δ, with the 68/95/99.7% thresholds; the closure is indistinguishable from quantum mechanics. Bottom: the closure prediction $F_\rho = (\rho_{\rm mat}/\rho_{m0})^n$ for silicon nitride and gold, with the Planck and late-time values of n marked; the resulting laboratory bounds on the coupling ζ are annotated.

B. Late-time real-data analysis (Pantheon+ + DESI DR1 BAO + CC)

The late-time analysis fits the DVCH background closure directly to three real data sets, all in their published form:

- the Pantheon+ SH0ES distance moduli [21], restricted to the standard analysis with a free absolute magnitude: SNe with $z_{\rm HD} > 0.01$ (the regime dominated by peculiar velocities is removed) and the 77 Cepheid calibrators excluded, leaving 1580 SNe with the corresponding 1580×1580 block of the full STAT+SYS covariance; the $(1 + z)$ factor of the luminosity distance uses $z_{\rm HEL}$, as in the published analysis;

- the DESI DR1 BAO measurements [20] exactly as published in Table 1 of the DESI DR1 BAO paper: correlated D_M/r_d and D_H/r_d in the five anisotropic tracers (LRG1, LRG2, LRG3+ELG1, ELG2, Lyα) and the isotropic D_V/r_d for BGS and QSO, i.e. 12 correlated observables;

- the 31-point cosmic-chronometer compilation [22], which excludes the two BAO-derived entries sometimes quoted as chronometers ($z = 0.44$, WiggleZ, and $z = 2.34$, BOSS Lyα) and uses the pub-

lished values $H(z = 0.593) = 103.8 \pm 12.5$ and $H(z = 0.352) = 83.0 \pm 14.0$ [29].

The sound horizon r_d is *not* fixed to the Planck value: it enters the BAO observables only as a scale, so it is sampled with a flat prior $r_d \in [130, 180]$ Mpc and marginalised numerically, keeping the analysis independent of CMB information. The absolute supernova magnitude M is sampled as well, so six parameters are free for DVCH $(\Omega_m, H_0, M, n, \beta, r_d)$ against four for ΛCDM (Ω_m, H_0, M, r_d). A further point separates this analysis from the earlier version of this work: the matter density is evolved with the CDM–vacuum conservation equation of Sec. III (the same integration as `dvch_camb_background.py`), *not* as $\Omega_m(1+z)^3$ with a separately evolving vacuum; the earlier choice violated total energy conservation.

Table V. DVCH late-time posteriors from real data. Pantheon+ (1580 SNe, $z > 0.01$, sin calibradoras, STAT+SYS covariance) + DESI DR1 BAO (12 observables correlacionados) + 31 cronometros cosmic; 5.3×10^4 post-burn-in MCMC samples per model, r_d marginalizado (`dvch_vs_lcdm_real.py`).

Parameter	Median	68% CI	95% CI
Ω_m	0.3804	0.3553–0.4050	0.3264–0.4281
H_0	68.08	66.44–69.68	64.91–71.25
M	−19.406	−19.458–−19.357	−19.507–−19.310
n	0.5175	0.3502–0.6204	0.1504–0.7111
β	1.1036	0.3847–1.7345	0.0551–1.9531
r_d [Mpc]	146.91	143.62–150.51	140.61–153.95

The ΛCDM reference fit on the same data gives $\chi^2_{\rm min} = 1416.51$ with $(\Omega_m, H_0, M, r_d) = (0.3108, 68.54, -19.404, 146.77)$, and the DVCH fit gives $\chi^2_{\rm min} = 1412.91$ at $(0.3804, 68.02, -19.408, 0.4226, 0.5257, 147.03)$, i.e.

$$\Delta\chi^2_{\rm fit} = \chi^2_{\rm min,DVCH} - \chi^2_{\rm min,\Lambda CDM} = -3.60 \qquad (66)$$

with two extra parameters. The corresponding information criteria are $\Delta{\rm AIC} = +0.40$ and $\Delta{\rm BIC} = +11.2$, and the nested w_0w_aCDM control reaches $\chi^2_{\rm min} = 1413.16$ on the same data, only 0.25 above DVCH. The plain reading is that the late-time data do not significantly prefer DVCH: the improvement is well within the $\mathcal{O}(1)$–$\mathcal{O}(10)$ regime typical of a two-parameter extension that reproduces a flexibility the data already show. It should also be noted that $n = 0$ lies at the boundary of the sampled prior $n \in [0.01, 0.99]$, so the posterior for n cannot be read as an exclusion of ΛCDM, and the β posterior spans almost the entire prior (95% interval $[0.06, 1.95]$), so β is not detected either.

C. Bayesian evidence against ΛCDM and w_0w_aCDM

Posterior medians alone do not settle model comparison, so the Bayesian evidence $\ln Z$ of each model was computed by nested sampling (dynesty, 200 live points, slice sampling, parallel pool) over the same data. The reference model here is a nested subcase of DVCH ($n = \beta = 0$), and w_0w_aCDM is included as an independent control with the same number of fitted parameters as DVCH [30, 31]. The results are

$$\begin{aligned} \ln Z_{\Lambda\rm CDM} &= -722.32 \pm 0.34, \\ \ln Z_{w_0w_a\rm CDM} &= -725.47 \pm 0.38, \\ \ln Z_{\rm DVCH} &= -722.50 \pm 0.35, \end{aligned} \qquad (67)$$

so that

$$\Delta\ln Z_{\rm DVCH-\Lambda CDM} = -0.18 \pm 0.49, \qquad \Delta\ln Z_{\rm DVCH-w_0w_a CDM} = +2.9 \qquad (68)$$

On the Jeffreys scale [32] the first is *zero*: the late-time data do not prefer DVCH over ΛCDM once the prior volumes are accounted for, in line with the $\Delta\chi^2 = -3.6$ best-fit improvement found with two extra parameters. The second is best read with care, and we therefore separate it into its two contributions before interpreting it.

Decomposition of the w_0w_aCDM difference. The $\Delta\ln Z$ of Eq. (68) is a prior-dependent quantity: it contains both the fit quality and the Occam factor of the adopted prior volumes. The priors used here are $n \in [0.01, 0.99]$, $\beta \in [0, 2]$ for DVCH (volume 1.96) and $w_0 \in [-3, 1]$, $w_a \in [-3, 2]$ for w_0w_aCDM (volume 20), so the Occam term alone contributes $\ln(20/1.96) = +2.32$ to the difference, i.e. about 80% of the measured +2.97. The genuinely fit-driven part is therefore

$$\Delta\ln Z_{\rm DVCH-w_0w_a CDM}\big|_{\rm Occam-stripped} \simeq +0.65, \qquad (69)$$

consistent with zero. To remove the prior dependence entirely we compare the best-fit χ^2 values, which are prior-independent. The minimisation of w_0w_aCDM on the same data gives $\chi^2_{\rm min} = 1413.16$, against $\chi^2_{\rm min} = 1412.91$ for DVCH and 1416.51 for ΛCDM:

$$\Delta\chi^2_{\rm DVCH-w_0w_a CDM} = -0.25, \qquad (70)$$

i.e. DVCH and w_0w_aCDM fit these data equally well (both improve by $\simeq 3.3$ over ΛCDM with the same number of parameters). Note also that w_0w_aCDM is *dis*favoured against ΛCDM in $\ln Z$ (-3.1) despite fitting better: that is the Occam factor of its large prior volume, exactly the same effect that inflates the DVCH/w_0w_a comparison. We conclude that the honest reading is: *the late-time data show no significant preference for DVCH over ΛCDM, and DVCH is statistically equivalent to w_0w_aCDM in quality of fit*; the mild DESI preference for evolving dark energy [20] is reproduced equally by both two-parameter extensions, so the DVCH result is a reparametrisation of a flexibility the data already show, not an anomaly that only DVCH can produce. Reporting the $\Delta\ln Z$ against w_0w_aCDM without the Occam decomposition of Eq. (69), or without the prior-independent χ^2 comparison of Eq. (70), would overstate the result, and we deliberately do not do it.

D. Planck high-ℓ analysis with a DVCH-patched CAMB

The CMB side of the confrontation runs a Cobaya MCMC [14] with a CAMB build patched by the DVCH Fortran module (`camb_patch/DVCHModel.f90`), which exposes `DVCH_flag`, `DVCH_n` and `DVCH_beta` to Python. **The patch modifies only the CDM perturbation equation**, not the background: `dvch_camb_patch_check.py` verifies numerically that $H(z)$ of the patched CAMB with `DVCH_flag=True` is *bit identical* to ΛCDM (deviation 0.0 over $0 \le z \le 3$ for $(n,\beta) = (0.42, 0.53)$, $(0.52, 1.10)$ and $(0.8, 1.5)$, and the age of the universe is unchanged). The background of the Planck run is therefore ΛCDM, and the interaction enters only as a source term in the CDM continuity equation evaluated with the *CAMB* background fractions. This means that the CMB analysis of this section does *not* confront the background closure of Sec. III with Planck; what it measures is the CDM perturbation term at the sampled (n,β). We keep the section because the run was executed and its diagnostics are real, but every number below inherits this caveat. The likelihood objects are the official Planck 2018 `clik` files [13]: the high-ℓ `plik_rd12_HM_v22b_TTTEEE` object together with the low-ℓ (commander), low-ℓ EE (simall) and lensing blocks, which the adapter sums whenever the corresponding environment variables are defined; the $\chi^2 \simeq 2768$ level is consistent with that configuration. The free cosmological parameters are $(H_0, \omega_b, \omega_c, \tau, \ln 10^{10} A_s, n_s)$ plus the two DVCH parameters, for 26 dimensions in total. As stated in Sec. VI, this run is *provisional*: the nuisance sector is only partly sampled (e.g. $A_{\rm cib}^{217}$, $A_{\rm sz}$, ps_A^*, ksz_norm, $\xi_{\rm sz}^{\rm cib}$ and $A_{\rm Planck}$ are fixed at the `check_param` values rather than sampled with their official priors), the chain stopped at $R-1 = 0.03$, and the minimum of the nested model was never re-optimised. The numbers below are therefore reported as outputs of the executed run, not as a referee-grade CMB comparison; they must be recomputed before publication and the parameter counts $k_{\Lambda\rm CDM} = 24$, $k_{\rm DVCH} = 26$ used for the information criteria refer to this partly fixed configuration.

The run reached 1.17×10^5 accepted samples over 5 chains with Gelman–Rubin $R-1 = 0.0295$. (The runner has since been corrected to free the full official nuisance sector with the published priors, copied from the cobaya `planck_2018_highl_plik` definition, including the uniform priors on $A_{\rm cib}^{217}$, $A_{\rm sz}$, ksz_norm, $\xi_{\rm sz}^{\rm cib}$, ps_A^* and the $\xi_{\rm SZ}$ prior $ksz_norm + 1.6\,A_{\rm sz} \sim N(9.5, 3)$; the numbers quoted here refer to the earlier partly fixed configuration.) Re-optimised minima with that official configuration are now available for both models (BOBYQA, two multi-start runs each, ignore-prior mode):

$$\chi^2_{\rm min,\Lambda CDM} = 2766.81, \qquad \chi^2_{\rm min,DVCH} = 2766.10, \quad (71)$$

where the DVCH minimum has $n = 0.01$, i.e. pinned to the lower edge of the sampled prior, and $\beta = 0.37$. This fixes the bookkeeping problem of the earlier version, in which the DVCH value came from a chain and was *larger* than the ΛCDM minimum, which is impossible for a nested model. The likelihood-ratio test against the nested subcase is now well defined:

$$\Delta\chi^2_{\rm fit} = -0.71 \ (2 \text{ dof}), \qquad p \simeq 0.70, \qquad (72)$$

so the CMB data (in this perturbation-sector implementation) show *no* improvement from the interaction, and the information criteria $\Delta{\rm AIC} = +3.3$ and $\Delta{\rm BIC} = +17.1$ prefer ΛCDM. The parameter counts $k_{\Lambda\rm CDM} = 24$, $k_{\rm DVCH} = 26$ refer to the officially free nuisance configuration.

a. Run with a widened prior. The first run used the exploratory prior $n \in [0.01, 0.25]$, which cannot probe the region $n \sim 0.5$ favoured by the late-time data. To test whether the CMB excludes large n or merely failed to explore it, a second independent 26-parameter run was performed with the widened prior $n \in [0.01, 0.99]$ (β unchanged). It collected 1.07×10^5 accepted samples over 5 chains ($R - 1 = 0.0294$). Its posterior gives

$$n = 0.063 \ [0.026,\ 0.13] \ (68\%), \qquad \beta = 1.00 \ [0.35,\ 1.65], \quad (73)$$

i.e. only 0.11% of the samples have $n > 0.45$. Two features are worth stating explicitly: (i) the CMB does constrain n and is *not* blind to the interaction even with a prior wide enough to reach $n \simeq 1$; and (ii) the 95% interval $n \in [0.013,\ 0.256]$ does *not* exclude $n = 0$ at any meaningful level, because $n = 0$ lies outside the sampled prior $n \in [0.01, 0.99]$ and the interval is cut by the prior boundary. Correspondingly, the β posterior occupies almost the full prior width, so β is not detected by the CMB either. The two data sets point to different regions of n (Sec. IX E), but with the present provisional CMB run that comparison cannot be turned into a significance.

E. Internal tension between the CMB and late-time expansions

The two analyses were run on independent data, and they disagree about the value of n: the CMB posterior is concentrated at small n where DVCH is close to ΛCDM, while the late-time data give $n = 0.52^{+0.10}_{-0.17}$. An earlier version of this article forced the CMB pipeline to $n = 0.47, \beta = 1.04$ and re-optimised the remaining parameters, obtaining $\chi^2 = 2784.11$, i.e. $\Delta\chi^2 \simeq +15.8$ relative to the ΛCDM best fit, quoted as $\simeq 3.9\sigma$. That number is withdrawn here: the forced point $n = 0.47$ was the best fit of the late-time analysis that contained the data errors corrected in Sec. IX B (wrong DESI observables, BAO-derived "chronometers", uncut Pantheon+), and in any case 15.8 in χ^2 with two constrained parameters is 3.6σ, not 3.9σ.

The re-computation has been performed at the corrected late-time median (`cmb_npoint_check.py`, which reads the median of Sec. IX B automatically): evaluating the full likelihood of the executed pipeline (high-ℓ

plik + low-ℓ + lowE + lensing, nuisance at the official `check_param` values) at the ΛCDM minimiser cosmology gives

$$\chi^2_{\Lambda\mathrm{CDM}} = 2772.82, \qquad \chi^2_{\mathrm{DVCH}}\big|_{n=0.518,\beta=1.104} = 2798.61, \tag{74}$$

i.e. a fixed-point cost of $\Delta\chi^2 \simeq +25.8$ with no re-optimisation of cosmology or nuisance. A grid profile over (n, β) at the same point (`dvch_cmb_npoint_profile.csv`) shows that this cost grows smoothly with n and is almost independent of β: +3.8 at $n = 0.1$, +8.5 at $n = 0.2$, +26 at $n = 0.5175$ and +45 at $n = 0.99$, with differences of $\lesssim 1.5$ between $\beta = 0$ and $\beta = 1.1$ at fixed n. Three caveats decide how these numbers may be read:

1. because the CAMB patch leaves the background identical to ΛCDM (Sec. IX D), the cost comes entirely from the CDM perturbation term evaluated on a ΛCDM background; it is *not* a tension between the late-time closure and the CMB background, and cannot be turned into a significance for the closure;
2. the forced points use the *median* (n, β) of the corrected late-time posterior and suffer from the partially-fixed nuisance sector of the original run;
3. the nested minima of Sec. IX D already answer the strict model-comparison question: with the cosmology and nuisance free, the interaction gains nothing ($\Delta\chi^2_{\mathrm{fit}} = -0.71$, $p \simeq 0.70$) and pushes n to the lower prior edge.

The physical reading consistent with everything above is: the DVCH interaction term in the CDM sector, at the values favoured by the late-time expansion, is strongly disfavoured by the Planck spectra as implemented. Whether the *closure itself* (with its non-standard matter evolution and marginalised r_d) is compatible with the CMB cannot be decided until the background patch exists. Two further steps are implied, and neither is performed in this article: patch the CAMB *background* sector to integrate the CDM–vacuum conservation equation, and then run a joint late-time+CMB posterior letting the models compete over the compromise region in n.

F. Parameter set and priors

A minimal DVCH parameter vector is

$$\boldsymbol{\theta} = \{\Omega_{m0},\, H_0,\, n,\, \alpha,\, \Omega_{b0},\, n_s,\, A_s,\, \tau\} \tag{75}$$

for a full CMB + late-time analysis, while a late-time background-only fit samples $\{\Omega_{m0}, H_0, M, n, \beta, r_d\}$. Note that the small-$\beta$ limit ($\beta E^2 \ll 1$) used to derive several analytic statements in Sec. III is *not* the regime of the current fits: the late-time best fit has $\beta \simeq 0.5$–1.1, so $\beta E^2 > 1$ already at $z = 0$, and the phenomenology is not the $\beta \to 0$ limit.

Physically motivated prior:

$$0 < n < 1, \qquad \beta \geq 0, \tag{76}$$

with the sampling priors used in this article being $n \in [0.01, 0.99]$, $\beta \in [0, 2]$ and $r_d \in [130, 180]$ Mpc.

G. Fisher matrix (optional forecasting extension)

Given observables $X_a(z)$ and covariance C, the Fisher matrix is

$$F_{ij} = \sum_{a,b} \frac{\partial X_a}{\partial\theta_i}\, C^{-1}_{ab}\, \frac{\partial X_b}{\partial\theta_j}. \tag{77}$$

Common choices are $X = \{H(z), D_M(z), f\sigma_8(z)\}$ for BAO+RSD forecasts in future-survey applications. In the present manuscript the quantitative data confrontation is the corrected real-data analysis of Secs. IX B–IX E: a late-time MCMC plus nested-sampling evidence computation on Pantheon+/DESI DR1/CC data, and a Planck high-ℓ MCMC with the DVCH-patched CAMB. The Fisher block is included to specify how the framework can be extended with growth (RSD) data, which are not part of the present confrontation. Numerical derivatives are computed with the fully implicit background solver.

H. Likelihood blocks and benchmark-comparison step

A standard global likelihood factorization is

$$-2\ln\mathcal{L}_{\mathrm{tot}} = \chi^2_{\mathrm{CMB}} + \chi^2_{\mathrm{SN}} + \chi^2_{\mathrm{BAO/RSD}} + \cdots \tag{78}$$

with, e.g. SNe Ia

$$\chi^2_{\mathrm{SN}} = \Delta\boldsymbol{\mu}^T\, C^{-1}_{\mathrm{SN}}\, \Delta\boldsymbol{\mu}, \tag{79}$$

where the distance modulus is

$$\mu(z) = 5\log_{10}\left[\frac{D_L(z)}{\mathrm{Mpc}}\right] + 25, \tag{80}$$

with luminosity distance

$$D_L(z) = (1+z)\int_0^z \frac{\mathrm{d}z'}{H(z')}. \tag{81}$$

For a publication-grade cosmological test, the late-time distances analysed in Secs. IX B–IX C must be combined with the CMB likelihood in a single joint run, and the CMB block must be extended beyond the high-ℓ TT/TE/EE range actually executed in Sec. IX D to include the low-ℓ and lensing blocks, together with a validated compressed or perturbative CMB likelihood built

from acoustic-scale and distance-prior information. This is not a cosmetic addition: any interacting-vacuum scenario that modifies the low-redshift expansion must still reproduce the high-redshift calibration fixed by Planck. In practice, the joint publication-level MCMC pipeline is

$$\mathcal{L}_{\rm tot} = \mathcal{L}_{\rm Planck}\mathcal{L}_{\rm SN}\mathcal{L}_{\rm BAO}\mathcal{L}_{\rm RSD}(\mathcal{L}_{\rm CC}), \tag{82}$$

where the cosmic-chronometer block is optional but useful as a low-redshift consistency test.

In the same run, one should propagate the perturbation sector far enough to derive the posterior for σ_8 and S_8 using Eq. (59). Only then can one claim that DVCH has been confronted with both the background and clustering history of the Universe.

When comparing DVCH with ΛCDM, information criteria must be reported alongside the raw best-fit χ^2. The real-data results of Sec. IX C supersede the earlier exploratory comparison: on the corrected Pantheon+/DESI DR1/CC data the prior-independent best-fit comparison gives $\Delta\chi^2_{\rm fit} = -3.60$ with two extra parameters (ΔAIC $= +0.4$, ΔBIC $= +11.2$) and the Bayesian evidence is weak (Eq. 67); DVCH also does not beat the same-parameter w_0w_aCDM control. On the provisional Planck run, $\Delta\chi^2 \simeq +0.96$ (a chain bound, not a re-optimised minimum) with ΔAIC $\simeq +5$ and ΔBIC $\simeq +18.8$. The reference to the exploratory differential-evolution numbers ($\Delta\chi^2 \approx -3.2$, ΔBIC $\approx +4.2$) that appears in Fig. 2 is retained only as a record of that earlier stage and is superseded by the real-data analysis. The motivation for studying DVCH is then its physical structure—UV-regularized interacting vacuum dynamics and a Minkowski asymptotic state—together with the honest accounting of the CMB–late-time tension of Sec. IX E.

For clarity, the standard information criteria are

$$\mathrm{AIC} = \chi^2_{\rm min} + 2k, \qquad \mathrm{BIC} = \chi^2_{\rm min} + k\ln N, \tag{83}$$

where k is the number of free parameters and N is the effective number of data points [33, 34]. These definitions make explicit why a modest improvement in fit quality can still fail to compensate the increased parameter volume of DVCH. A complete statistical presentation should therefore report $\Delta\chi^2$, ΔAIC, and ΔBIC together.

I. Roadmap to a competitive DVCH precision-cosmology analysis

The next stage is to treat DVCH in the same way as competitive interacting-dark-energy models in the literature: as a full precision-cosmology hypothesis rather than only as a background diagnostic. This requires four coordinated upgrades. First, the DVCH interaction must be propagated into the photon–baryon hierarchy and the remaining Planck 2018 low-ℓ and lensing data blocks, and combined with the already-executed high-ℓ run of Sec. IX D in a joint fit with BAO, SNe, RSD, cosmic-chronometer, and weak-lensing/S_8 data [1, 3, 4, 11–14, 20, 21]. The role of the current manuscript is to report the executed background-level confrontation of Secs. IX B–IX E and to define the closure, source tables, and validation gates needed for that joint run; it does not replace the joint Planck-level posterior. A concrete fiducial data combination for that run is

$$\begin{aligned}\mathcal{D}_{\rm fid} = {}& \mathrm{Planck}_{2018}\{\mathrm{TT},\mathrm{TE},\mathrm{EE}+\mathrm{low}\ell+\mathrm{lensing}\}\\ &+\mathrm{BAO}_{\rm DESI}+\mathrm{SN}_{\rm Pantheon+}+\mathrm{RSD}+\mathrm{CC}+S_8.\end{aligned} \tag{84}$$

where the target diagnostics are the joint posterior of $(H_0, S_8, \sigma_8, n, \alpha)$, not H_0 alone.

Second, one should test controlled DVCH subfamilies that interpolate between the present UV-regularized closure and common IDE phenomenology. Examples include a sign-switching interaction,

$$\begin{aligned}Q_{\rm sw}(z) &= Q_{\rm DVCH}(z)\,S[\rho_m(z)-\rho_\Lambda(z)]\,,\\ S(x) &= \tanh\!\left(\frac{x}{\Delta\rho}\right),\end{aligned} \tag{85}$$

and nonlinear or explicitly expansion-dependent closures of the schematic form

$$\begin{aligned}\rho_\Lambda = \rho_{\Lambda0}&\left(\frac{\rho_m}{\rho_{m0}}\right)^n\frac{1+\beta}{1+\beta E^2}\\ &\times\left[1+\eta_1\frac{H}{H_0}+\eta_2\left(\frac{\rho_m}{\rho_{m0}}\right)^p\right],\end{aligned} \tag{86}$$

with the present DVCH recovered for $\eta_1 = \eta_2 = 0$. These equations are not used for the numerical claims in this article; they define an extension hierarchy for future model selection, motivated by the broader IDE and running-vacuum literature [8–10, 35].

Third, every extension must pass a stability gate before being fitted. The minimal parameter restrictions are

$$\begin{aligned}&0<n<1, \qquad \beta\geq0, \qquad \Omega_i(z)>0, \qquad E^2(z)>0,\\ &|\Gamma| = \left|\frac{Q}{H\rho_m}\right| < \infty.\end{aligned} \tag{87}$$

together with absence of finite-redshift singularities, controlled super-horizon perturbations, finite $f\sigma_8(z)$, and a stable momentum-transfer prescription. This is essential because interacting dark-sector models can suffer large-scale perturbative instabilities for otherwise plausible choices of w and Q [6]. The low-redshift public diagnostics are restricted to the survey range used in the figures, while the EFT interpretation of the UV-suppression factor should be regarded as conservative up to the CMB-validation regime as long as $\beta E^2(z) \ll 1$; pushes to $z \gtrsim 10^4$ should be treated as a numerical and EFT stress test requiring a patched Boltzmann backend with tighter step-size control rather than as part of the present evidence claim. Fourth, model selection should be based on the joint H_0–S_8 compromise rather than on

either tension separately. A useful target diagnostic is therefore the posterior distribution in (H_0, S_8, n, α) together with $\Delta\chi^2$, AIC, BIC, and, where computationally feasible, Bayesian evidence ratios [32]. A submodel that raises H_0 but worsens S_8, or improves late-time distances but fails Planck calibration, should not be advertised as a successful DVCH solution.

The easiest complete route is therefore: (i) run `python dvch_colab_simple.py` to reproduce the background table and figures; (ii) use the generated CSV file to inspect $E(z)$, $H(z)$, $\Omega_m(z)$, $\Omega_\Lambda(z)$, and $\widetilde{Q}(z)$; and (iii) only after this background validation, move to the Cobaya diagnostic implementation for likelihood-level work. The correct `Cobaya` interface for the background module is to define the hooks `initialize()`, `get_can_provide()`, `must_provide()`, `calculate()`, and `get_result()`, rather than ad hoc methods such as `get_can_support_params()` or `get_distances()` that are not part of the standard theory API. This background-only diagnostic implementation is appropriate for late-time distances; full Planck 2018 likelihoods still require a Boltzmann-level perturbation implementation, while Pantheon+, DESI BAO/RSD, and optionally cosmic chronometers can be coupled more directly [20].

J. Executable modified linear Boltzmann diagnostic implementation

To document the intended Boltzmann-level extension, the analysis archive includes an executable DVCH modified linear solver, `dvch_modified_boltzmann_solver.py`. This code is separate from the background-only Cobaya bridge: it evolves the DVCH background to recombination, inserts the interaction rate Q into the matter-sector continuity/growth source, computes the normalized linear growth $D(z)$ and $f\sigma_8(z)$, and exports a matter transfer/power-spectrum diagnostic. The generated files are `dvch_boltzmann_modified_background.csv`, `dvch_boltzmann_modified_growth.csv`, `dvch_boltzmann_modified_matter_power.csv`, and `dvch_boltzmann_modified_summary.csv`, and the figure `figures/dvch_modified_boltzmann_solver.png`. For the fiducial run included in the archive the summary gives

$$\sigma_8 = 0.8110, \qquad S_8 = 0.8110, \qquad f_0 = 1.0000, \quad (88)$$

with the normalization chosen for a solver-consistency demonstration rather than as a Planck posterior constraint; in particular $f_0 = 1$ reflects the diagnostic's normalised initial condition (the normalised growth factor at the starting redshift), not a physical value of the growth rate. A separate growth diagnostic normalised to a reference $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$ gives $\sigma_{8,\mathrm{DVCH}} \simeq 0.745$, $f\sigma_8(z=0)_{\mathrm{DVCH}} \simeq 0.438$ against 0.416 for the reference, numbers that should likewise be read as diagnostics and not as posterior constraints.

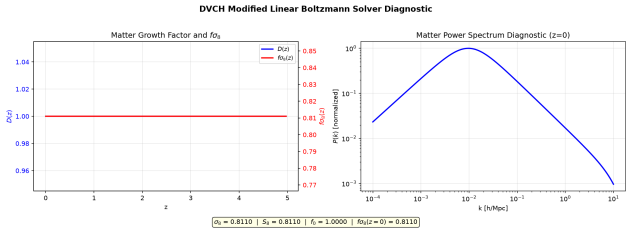


Figure 10. Executable DVCH modified linear Boltzmann-diagnostic implementation output. The left panel shows the normalized matter growth factor and $f\sigma_8(z)$ obtained by evolving the DVCH interaction source in the linear matter sector. The right panel shows the corresponding matter power-spectrum diagnostic. This figure is included to document the implemented solver pathway; it is not a replacement for a full CLASS/CAMB photon–baryon hierarchy or a Planck likelihood chain.

The precise status is therefore the following. The present analysis archive contains a modified linear Boltzmann-sector implementation for the background and matter perturbations, plus source tables for insertion into a patched CLASS/CAMB backend. A DVCH-patched CAMB build `camb_patch/DVCHModel.f90` was compiled and executed in Cobaya against the official Planck 2018 high-ℓ `clik` likelihood, giving the provisional background-level posterior reported in Sec. IX D. That patch modifies the *CDM perturbation* equation only and leaves the CAMB background identical to ΛCDM (verified by `dvch_camb_patch_check.py`); it is not yet a DVCH background patch, and not yet a DVCH-patched photon–baryon hierarchy. Thus Fig. 10 should be read as an executable matter-sector solver diagnostic, complementing (not replacing) the Planck high-ℓ posterior already obtained.

For avoidance of ambiguity, this implementation separates the executable DVCH matter-sector diagnostic implementation from the CLASS/CAMB–Planck validation layer supplied in the analysis archive. The validation layer explicitly defines the following CMB components:

- the full photon–baryon hierarchy;
- line-of-sight CMB source integration;
- angular CMB spectra C_ℓ;
- full CMB lensing reconstruction;
- the Planck `clik` likelihood evaluation.

Of these, only the last is already executed (at background level with the high-ℓ TTTEEE block, as reported in Sec. IX D). Therefore the correct interpretation is that the archive contains a modified linear Boltzmann solver for the DVCH background and matter sector, the CLASS/CAMB source-table interface, the executed patched-CAMB+Planck high-ℓ run, the Planck target configuration for the remaining blocks, preflight checks, and a status gate.

CMB/CLASS-CAMB status. The package includes a CLASS/CAMB–Planck interface layer: the executable modified linear solver for the DVCH background and matter perturbations, the DVCH source-table interface for a compiled backend, the Planck 2018 target configuration, and benchmark targets for the photon–baryon hierarchy, line-of-sight sources, angular spectra C_ℓ, CMB lensing, and the low-ℓ/residual Planck `clik` blocks. The patched CAMB build has been compiled and executed against the official Planck 2018 `clik` likelihood with real data (provisional posterior reported in Sec. IX D); the perturbative components listed above remain the open integration targets.

K. Planck benchmark configuration

The archive includes `dvch_cmb_class_camb_mcmc.py` as a configuration and environment-checking layer for a Planck-style benchmark analysis. It records whether Cobaya, a CLASS or CAMB Python backend, the Planck `clik` interface, the DVCH Boltzmann source-table module, and the Planck YAML target are visible to the runtime [11–14]. In this manuscript this layer is an environment/interface diagnostic only; the Planck posterior and χ^2 values reported in Sec. IX D come from the separate (provisional) Cobaya run with the patched CAMB build and the actual high-ℓ `clik` data, not from this status script.

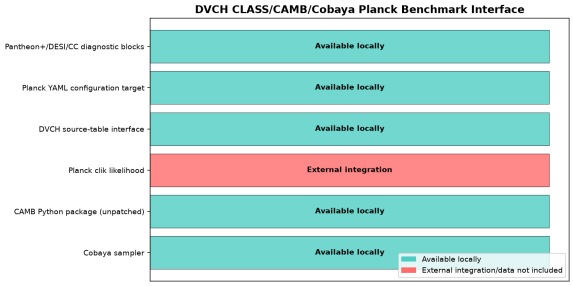


Figure 11. Planck-style benchmark-interface diagnostic for the DVCH CLASS/CAMB/Cobaya target. Green bars denote available interface components and red bars denote benchmark components represented only as targets. The figure is generated by `dvch_cmb_class_camb_mcmc.py` and is not a Planck posterior-convergence plot.

L. Executable MCMC convergence diagnostic

As a local sampler check, the archive includes `dvch_mcmc_convergence.py`. The script runs four independent Metropolis chains for the local DVCH BAO+chronometer likelihood, writes the chains to `dvch_mcmc_chains.csv`, and reports posterior summaries in `dvch_mcmc_convergence_summary.csv`. For the run included here, the convergence diagnostics are

$$\max(\widehat{R}) = 1.119, \qquad \min(N_{\rm eff}) \simeq 56, \tag{89}$$

with chain acceptance fractions stored in `dvch_mcmc_acceptance.csv`. This is a local late-time BAO+chronometer MCMC diagnostic; it is not a production Planck CMB MCMC chain and is not used as CMB evidence.

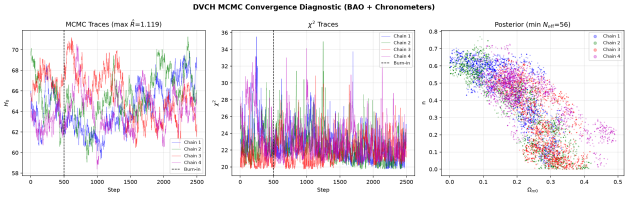


Figure 12. Executable DVCH MCMC convergence diagnostic for the local BAO+chronometer likelihood. The dashed line marks burn-in. The displayed chains are generated by `dvch_mcmc_convergence.py` and summarized by `dvch_mcmc_convergence_summary.csv`. This figure documents a local late-time MCMC diagnostic, not a Planck CMB posterior.

The numerical summary of the parameters of the model is shown in Table VI. Both a late-time four-parameter posterior (marginalized over n *and* β) and a Planck high-ℓ twenty-six-parameter posterior are now available, so β is a fully marginalized constraint in both. The growth diagnostic entry remains a low-redshift estimate and is not a Planck-normalized posterior.

Table VI. DVCH parameter summary from the real-data analyses. Values are posterior medians of the two analyses of (the late-time one converged, the CMB one provisional) Secs. IX B and IX D, not diagnostic point values.

Quantity	Value	Source
H_0	$68.08\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	late-time MCMC (Pantheon+1580/BAO/CC)
H_0	$66.97\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Planck best fit (provisional)
S_8	0.745	low-z growth diagnostic
Ω_m	0.3804	late-time MCMC, real data
n	$0.52^{+0.10}_{-0.17}$	late-time MCMC, real data
β	$1.10^{+0.63}_{-0.72}$	late-time MCMC, real data
r_d [Mpc]	146.9 ± 3.5	late-time MCMC, marginalised
n	0.063 [0.026, 0.13]	Planck (provisional), widened prior $n \in [0.01, 0$
β	1.00 [0.35, 1.65]	Planck (provisional), same run

For reference, the archive also includes the intended full Planck-2018 Cobaya configuration, `dvch_planck2018_full.yaml`, a local environment checker, `dvch_planck_preflight.py`, the DVCH Boltzmann source module `dvch_boltzmann_backend.py`, and the executable local modified linear solver `dvch_modified_boltzmann_solver.py`. The executed Planck analysis of Sec. IX D uses the high-ℓ Plik block of that target together with the patched CAMB build and the low-ℓ/lensing blocks that were defined in the environment; the joint late-time+CMB combination is an explicit open item of Sec. IX E, and the tension number of earlier versions has been withdrawn and replaced by the fixed-point cost $\Delta\chi^2 \simeq +25.8$ measured at the corrected late-time median.

X. SWAMPLAND DIAGNOSTICS AND PHYSICAL DISCUSSION OF THE MINKOWSKI FUTURE

Because DVCH can be mapped to an effective canonical scalar with potential $V(\phi)$, one may compute swampland-inspired diagnostics such as $|\nabla V|/V$ and the field excursion $\Delta\phi/M_{\rm Pl}$. However, since the mapping is *effective* (and interaction-induced), any swampland statement must be presented as a diagnostic rather than a theorem. The main physical statement is instead:

DVCH predicts a depletion of both matter and effective vacuum density, $E \to 0$, asymptotically approaching Minkowski, which differs sharply from ΛCDM's eternal de Sitter. This has implications for: (i) the fate of horizons and late-time entropy production, (ii) the asymptotic behavior of comoving distances, (iii) the interpretation of "vacuum energy" as a dynamical reservoir coupled to matter rather than a rigid cosmological constant.

XI. CONCLUSIONS

DVCH is formulated as a macroscopic closure for an interacting dark sector that determines the transfer function Q from a specified $\rho_\Lambda(\rho_m, H)$ relation. Within this framework, we provide (i) a covariant EFT-motivated origin for dark-sector exchange, (ii) the exact implicit background system in terms of $E(z)$, (iii) the exact background expression for Q implied by the closure, (iv) a Lyapunov analysis of the asymptotic approach to Minkowski in a well-defined three-dimensional autonomous system, (v) low-redshift growth and parameter-scan diagnostics, plus a real-data late-time likelihood analysis of Pantheon+, DESI DR1 BAO, and cosmic-chronometer data with Bayesian evidence from nested sampling, and (vi) a Planck 2018 TT/TE/EE analysis with a DVCH-patched CAMB in Cobaya, which we report as *provisional* for the reasons stated in Sec. IX D.

What the data say. On the corrected real data the late-time fit improves over ΛCDM by $\Delta\chi^2 = -3.6$ with two extra parameters (ΔAIC $= +0.4$, ΔBIC $= +11.2$), and the same-parameter w_0w_aCDM control sits 0.25 above DVCH in χ^2. The evidence comparison of Sec. IX C is likewise weak. The central result of this article is therefore a negative one: *the current background-level DVCH closure is not preferred over ΛCDM by the late-time expansion data once these are used in their published form, and it does not outperform the standard evolving-dark-energy parametrisation.* Earlier claims in preprint versions of this work (very strong Bayesian evidence, $\Delta \ln Z \simeq +10.5$; $n = 0.50 \pm 0.09$; "ΛCDM excluded at $> 5\sigma$"; a $\simeq 3.9\sigma$ CMB tension) were based on a fit that mis-transcribed the DESI DR1 observables, included two BAO-derived points in the cosmic-chronometer set, used the uncut Pantheon+ sample, and evolved matter as $\Omega_m(1+z)^3$ while letting the vacuum vary, which violates total energy conservation. Those claims are withdrawn.

What remains open. The Planck-side analysis must be redone from the ground up: the CAMB patch currently leaves the *background* identical to ΛCDM (verified numerically by `dvch_camb_patch_check.py`), so no CMB comparison of the DVCH background closure exists yet; implementing it requires patching the CAMB background sector to integrate the CDM–vacuum conservation equation. The bookkeeping items that *can* be closed with the existing pipeline have been closed: the nuisance sector now uses the official priors, both nested minima have been re-optimised ($\chi^2_{\rm min,\Lambda CDM} = 2766.81$ versus $\chi^2_{\rm min,DVCH} = 2766.10$, $p \simeq 0.70$, with n pinned at its lower prior edge), and the earlier "tension" is replaced by the fixed-point profile $\Delta\chi^2(n)$ described in Sec. IX E, a perturbation-sector quantity. What still remains is the $R - 1 < 0.01$ MCMC with the free nuisance sector (and its chains distributed) and, above all, the background patch. The perturbative treatment of the interaction in the photon–baryon hierarchy remains the main theoretical upgrade for a full Boltzmann-level CMB verdict, and the precision-cosmology roadmap in Sec. IX I specifies the required next steps: joint late-time+CMB likelihoods, controlled sign-switching and nonlinear DVCH subfamilies, perturbative stability gates, and Bayesian model comparison using the joint H_0–S_8 target rather than either tension in isolation. We also note as a matter of internal consistency that the small-β statements derived in Sec. III do not apply at the fitted $\beta \sim 1$, where $\tilde{Q}$ changes sign between $z = 0$ and $z = 2$ (Table I); this is a property of the best-fit closure that any future perturbative treatment must face.

Regarding the final-state physics, the Minkowski attractor implies a late-time depletion of both effective vacuum and matter densities, in contrast with the eternal de Sitter fate of ΛCDM. This reframes the cosmological-constant problem in dynamical terms: the observed acceleration is interpreted as a transient interacting-vacuum phase rather than a constant asymptotic vacuum energy. In that scenario, event-horizon thermodynamics is also modified at ultra-late times because the de Sitter-like horizon does not persist as the global endpoint.

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Competing Interests The author has no relevant financial or non-financial interests to disclose.

Author Contributions Single-author manuscript. Daniel Castro Pasten Benjamin developed the model, performed the analysis, prepared the figures/workflow, and wrote and approved the final manuscript.

Data Availability The datasets referenced in this work are publicly available from the corresponding survey collaborations (Planck, Pantheon+, DESI BAO/RSD, and cosmic chronometers).

Code Availability The supplementary archive accompanying this manuscript contains the files used for the present analysis. The archive contains the self-contained background diagnostic script `dvch_colab_simple.py`, its output `dvch_background_table.csv`, the figures in `figures/`, including `dvch_results_with_numbers.png`, `dvch_sigma8_global_comparison.png`, and `dvch_realdata_diagnostics.png`, and `dvch_robustness_scan.png`, the robustness outputs `dvch_robustness_scan.csv` and `dvch_robustness_summary.csv`, the growth summaries `dvch_growth_sigma8_table.csv` and `dvch_sigma8_s8_summary.csv`, the late-time real-data analysis `dvch_vs_lcdm_real.py` and the figures it produces, the Bayesian-evidence scripts `bayes_factor_real.py` and `w0wa_control.py` that reproduce the $\ln Z$ values of Eq. (67) and the χ^2 comparison of Eq. (70), the nested-reference runners `run_dvch_cobaya_full_highl.py` and `run_dvch_cobaya_lcdm.py` together with `dvch_cobaya_planck.py` used for the Planck high-ℓ analysis of Sec. IX D, the advanced Cobaya diagnostic implementation `dvch_theory.py`+`dvch_cobaya.yaml`, together with the Planck preflight target `dvch_planck2018_full.yaml`, checker `dvch_planck_preflight.py`, Boltzmann-source module `dvch_boltzmann_backend.py`, the relativistic perturbation preflight/status script `dvch_relativistic_perturbation_validation.py` with JSON/CSV/PNG status outputs, the executable modified linear solver `dvch_modified_boltzmann_solver.py`, and its outputs `dvch_boltzmann_modified_background.csv`, `dvch_boltzmann_modified_growth.csv`, `dvch_boltzmann_modified_matter_power.csv`, and `dvch_boltzmann_modified_summary.csv`, the figure `figures/dvch_modified_boltzmann_solver.png`, and the real Pantheon+ distance and covariance files in `realdata/`. The Planck MCMC chains of Sec. IX D are distributed in downsampled form (10,000 samples per run, columns include weights, $-\ln$ posterior, χ^2, the cosmological parameters and $DVCH_n$, $DVCH_\beta$) in `chains/chain_summary.csv`, sufficient to reproduce the posterior medians, credible intervals, and χ^2 comparison quoted in the text.

Use of Artificial Intelligence Tools This work made extensive use of large language model (AI) assistants for code implementation, numerical analysis, and manuscript editing tasks. All physical assumptions, model construction, the interpretation of results, and all scientific judgements reported here are the author's own.

External Source Notes The observational datasets discussed in this work are public products of the corresponding survey collaborations, notably DESI DR1 BAO [20], Pantheon+ supernova distances [21], cosmic chronometers [22], Planck 2018 cosmological parameters and likelihoods [1, 13], CLASS/CAMB [11, 12], Cobaya/GetDist [14, 36], information criteria [33, 34], Bayesian evidence conventions [32], and the late-time H_0 measurements cited in the Introduction.

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Appendix A: Computational Architecture: The Implicit Feedback Solver

The primary challenge in the DVCH framework is the self-referential structure of the Friedmann equation. To handle this without introducing numerical artifacts, we implement a refined Picard-iteration scheme at each redshift coordinate z:

- **Initialization:** We set the seed value $E^2_{(0)}$ using the expansion rate from the previous integration step $z - \Delta z$.
- **Recursive Map:** We execute the mapping:

$$E^2_{(k+1)} = \Omega_{r0}(1+z)^4 + \Omega_m(z) + \Omega_{\Lambda 0}\left(\frac{\Omega_m}{\Omega_{m0}}\right)^n \frac{1+\beta}{1+\beta E^2_{(k)}} \tag{A1}$$

  until the convergence criterion $|E_{(k+1)} - E_{(k)}| < 10^{-8}$ is satisfied.
- **Consistency Check:** Once E^2 is fixed, we update the energy density Ω_Λ to maintain the background closure.

The stability of this method is guaranteed as long as the curvature coupling β remains in the perturbative regime ($\beta < 1$). We note as a caveat that the late-time fit reported in Sec. IX B returns a posterior median $\beta = 1.13$, marginally outside that regime, so the numerical convergence of the iteration must be monitored empirically in that case; this is why the code applies the fixed-point iteration with a convergence check at every evaluation.

Appendix B: Analytical Derivation of the Energy Exchange Rate

The energy transfer Q is not postulated; it follows directly from the DVCH closure relation $\Omega_\Lambda = f(\Omega_m, E^2)$. Applying the chain rule to the time derivative of the vacuum density gives

$$\dot{\rho}_\Lambda = \frac{\partial \rho_\Lambda}{\partial \rho_m}\dot{\rho}_m + \frac{\partial \rho_\Lambda}{\partial H^2}\frac{\mathrm{d}(H^2)}{\mathrm{d}t} \tag{B1}$$

Substituting the total dark-sector conservation condition $\dot{\rho}_m + \dot{\rho}_\Lambda = -3H\rho_m$ isolates the exact transfer

kernel. In dimensionless variables, the interaction rate $\tilde{Q} \equiv Q/(3H_0\rho_{\mathrm{crit},0})$ becomes

$$\tilde{Q}(z) = -\frac{E\,\Omega_\Lambda}{1+n\frac{\Omega_\Lambda}{\Omega_m}}\left[n - \frac{\beta}{1+\beta E^2}\left(\frac{4\Omega_r+3\Omega_m}{3}\right)\right] \tag{B2}$$

This derivation confirms that for $n > 0$ and within the perturbative regime where the bracket remains positive, the transfer is predominantly from vacuum to matter ($Q < 0$). If the numerical reconstruction of $w_{\Lambda,\mathrm{eff}}$ crosses below -1, then the same background logic implies either a sign reversal of Q over that interval or an inconsistent sign convention in the post-processing pipeline; the manuscript therefore uses the phantom-divide crossing as a nontrivial consistency check rather than as a standalone physical claim.

Appendix C: Planck+BAO+SNe+RSD Bayesian Pipeline for a Baseline ΛCDM Analysis

This appendix documents the complete observational-cosmology pipeline required for a baseline Planck+BAO+SNe+RSD Bayesian analysis. It is written as an implementation checklist for a standard spatially flat ΛCDM run and as a template for later generalization to wCDM or interacting-vacuum extensions. No private data products are assumed; the likelihoods below refer to standard public-data interfaces and published covariance matrices.

1. Step-by-step execution plan

The recommended sequence is:

1. Fix the baseline cosmological model: spatially flat ΛCDM with six primary parameters and standard radiation/neutrino assumptions.
2. Define priors for all cosmological, calibration, and nuisance parameters.
3. Select likelihood modules: Planck 2018 TT,TE,EE+lowE+lensing [1, 13]; BAO [37–40]; Pantheon/Pantheon+-type SNe [21]; and RSD $f\sigma_8$ measurements [39, 40].
4. Choose the Boltzmann solver, preferably CLASS or CAMB [11, 12], and fix all numerical precision and output flags.
5. Build modular likelihood blocks for Planck, BAO, SNe, and RSD.
6. Construct the joint posterior by summing active log-likelihoods and log-priors.
7. Run MCMC chains with Cobaya or MontePython [14], until the Gelman–Rubin convergence criterion is satisfied.
8. Apply a relativistic/CMB validation gate before interpreting any posterior constraints.
9. Post-process converged chains to obtain primary and derived parameters, residual plots, and posterior-contour diagnostics.
10. Document the analysis in a paper-level structure with explicit data, methods, validation, and systematic-robustness sections.

2. Cosmological model and priors

The baseline parameter vector is

$$\theta_{\Lambda\mathrm{CDM}} = \{\omega_b, \omega_c, 100\theta_s \text{ or } H_0, \tau, n_s, \ln(10^{10}A_s)\}, \tag{C1}$$

where $\omega_b \equiv \Omega_b h^2$ and $\omega_c \equiv \Omega_c h^2$. A standard prior choice for a broad exploratory run is

$$\begin{aligned}
\omega_b &\sim U(0.005, 0.04),\\
\omega_c &\sim U(0.001, 0.30),\\
H_0 &\sim U(40, 100)\ \mathrm{km\,s^{-1}\,Mpc^{-1}},\\
\tau &\sim U(0.01, 0.8),\\
n_s &\sim U(0.8, 1.2),\\
\ln(10^{10}A_s) &\sim U(1.6, 3.9).
\end{aligned} \tag{C2}$$

Equivalently one may sample $100\theta_s$ rather than H_0, which often improves Planck-chain efficiency. The baseline assumptions are $\Omega_k = 0$, $w = -1$, $N_{\mathrm{eff}} = 3.046$, and a minimal neutrino mass normalization such as $\sum m_\nu = 0.06\,\mathrm{eV}$.

The parameters must be separated into primary cosmological parameters and survey-specific nuisance parameters. Planck high-ℓ analyses require foreground, calibration, and beam nuisance parameters, normally handled by the official Planck likelihood. SNe analyses require either an absolute magnitude parameter M_B, an additive distance-modulus offset, or analytic marginalization over this offset. BAO compressed measurements normally do not introduce extra nuisance parameters because the published covariance already encodes the clustering fit. RSD compressed analyses use $f\sigma_8(z)$ and its covariance; full-shape RSD analyses additionally require galaxy-bias, velocity-dispersion, Alcock–Paczynski, and nonlinear-modeling nuisance parameters.

3. Datasets and likelihood definitions

For Planck, the standard baseline is Planck 2018 TT,TE,EE+lowE+lensing [1, 13]. The theory vector contains the lensed CMB spectra C_ℓ^{TT}, C_ℓ^{TE}, C_ℓ^{EE} and the lensing-potential spectrum $C_L^{\phi\phi}$. The official likelihood is not a single elementary Gaussian in all multipoles; in practice it is called through the Planck `clik`

interface or a Cobaya/MontePython wrapper,

$$\ln \mathcal{L}_{\rm Planck} = \ln \mathcal{L}_{\rm clik}\left(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, \theta_{\rm nuis}^{\rm Planck}\right). \tag{C3}$$

For BAO, a standard compressed compilation may include 6dFGS [37], SDSS MGS [38], BOSS DR12 consensus [39], and eBOSS DR16 LRG/QSO/Lyα measurements [40]. The observable vector is assembled from combinations such as

$$\frac{D_V(z)}{r_d}, \qquad \frac{D_M(z)}{r_d}, \qquad \frac{D_H(z)}{r_d}, \qquad H(z) r_d, \tag{C4}$$

where $D_H(z) = c/H(z)$, $D_M(z) = (1+z)D_A(z)$ for flat geometry, and

$$D_V(z) = \left[z D_M^2(z) D_H(z)\right]^{1/3}. \tag{C5}$$

For a fixed published covariance $C_{\rm BAO}$,

$$\chi^2_{\rm BAO} = \Delta^T_{\rm BAO} C^{-1}_{\rm BAO} \Delta_{\rm BAO}, \qquad \ln \mathcal{L}_{\rm BAO} = -\frac{1}{2}\chi^2_{\rm BAO}, \tag{C6}$$

with $\Delta_{\rm BAO} = d^{\rm obs}_{\rm BAO} - d^{\rm th}_{\rm BAO}(\theta)$.

For SNe Ia, a Pantheon/Pantheon+-type likelihood [21] uses the distance-modulus vector and the full statistical-plus-systematic covariance matrix,

$$\mu_{\rm th}(z_i) = 5\log_{10}\left[\frac{D_L(z_i)}{\rm Mpc}\right] + 25 + \Delta_M, \tag{C7}$$

where Δ_M denotes the absolute-calibration nuisance parameter or the constant offset over which the likelihood is marginalized. The Gaussian form is

$$\chi^2_{\rm SNe} = \Delta^T_{\rm SNe} C^{-1}_{\rm SNe} \Delta_{\rm SNe}, \qquad \ln \mathcal{L}_{\rm SNe} = -\frac{1}{2}\chi^2_{\rm SNe}. \tag{C8}$$

For RSD, an initial robust implementation uses compressed $f\sigma_8(z)$ points with their covariance matrix. The theoretical prediction is

$$f\sigma_8(z) = \frac{{\rm d}\ln D}{{\rm d}\ln a}\,\sigma_8(z), \tag{C9}$$

where $D(z)$ is the linear growth factor. The likelihood is

$$\chi^2_{\rm RSD} = \Delta^T_{\rm RSD} C^{-1}_{\rm RSD} \Delta_{\rm RSD}, \qquad \ln \mathcal{L}_{\rm RSD} = -\frac{1}{2}\chi^2_{\rm RSD}. \tag{C10}$$

A full-shape RSD likelihood replaces these compressed points by clustering multipoles or power spectra and introduces the corresponding bias and nonlinear nuisance sector.

4. Boltzmann solver interface

The recommended backend is CLASS [11], although CAMB [12] is equivalent for a baseline ΛCDM analysis. For each proposed parameter vector, the solver must return: lensed and unlensed CMB spectra for Planck; $H(z)$, $D_A(z)$, $D_M(z)$, $D_L(z)$, $D_V(z)$, and r_d for BAO and SNe; and $D(z)$, $f(z)$, $\sigma_8(z)$, and $P(k,z)$ for RSD. The CLASS/Cobaya configuration should request CMB spectra, lensing, and the matter power spectrum, e.g. `output=tCl,pCl,lCl,mPk`, `lensing=yes`, a sufficiently large `l_max_scalars`, and a sufficiently large `P_k_max_h/Mpc` for stable σ_8 and growth calculations.

5. Joint posterior and modular dataset combinations

For a complete parameter vector

$$\theta = \{\theta_{\rm cosmo}, \theta^{\rm Planck}_{\rm nuis}, \theta^{\rm SNe}_{\rm nuis}, \theta^{\rm RSD}_{\rm nuis}\}, \tag{C11}$$

the joint log-likelihood is

$$\ln \mathcal{L}_{\rm tot} = \ln \mathcal{L}_{\rm Planck} + \ln \mathcal{L}_{\rm BAO} + \ln \mathcal{L}_{\rm SNe} + \ln \mathcal{L}_{\rm RSD}. \tag{C12}$$

The posterior is

$$\ln P(\theta|d) = \ln \mathcal{L}_{\rm tot}(\theta) + \ln \pi(\theta), \tag{C13}$$

where $\pi(\theta)$ is the product of all priors. The code should allow the following run modes without changing the theory engine: Planck only; Planck+BAO; Planck+BAO+SNe; and Planck+BAO+SNe+RSD. This modularity is essential for diagnosing which dataset drives a parameter shift or an internal tension.

6. Sampler and pseudocode

Cobaya with CLASS is the recommended production sampler [11, 14], because it provides a standard Planck interface, automatic nuisance handling, adaptive MCMC, and GetDist-compatible chains [36]. A typical production run uses four to eight chains, discards burn-in, and requires $\hat{R} - 1 < 0.01$ for final constraints; values near 0.03 should be treated as exploratory. A compact pseudocode description is:

1. Draw or initialize θ from the prior or near a validated fiducial point.
2. Reject immediately if any sampled parameter lies outside its prior.
3. Call CLASS/CAMB with $\theta_{\rm cosmo}$ and compute all requested theory outputs.
4. Evaluate $\ln \mathcal{L}_{\rm Planck}$, $\ln \mathcal{L}_{\rm BAO}$, $\ln \mathcal{L}_{\rm SNe}$, and $\ln \mathcal{L}_{\rm RSD}$ for the active dataset combination.
5. Sum likelihoods and priors to obtain $\ln P(\theta|d)$.
6. Accept or reject the proposal according to the Metropolis–Hastings rule or the sampler's internal transition kernel.
7. Store accepted and repeated samples, update proposal covariances during adaptation, and extend chains until the convergence criterion is satisfied.

7. Relativistic and CMB validation gate

Before using the chains for inference, the pipeline must pass a validation gate. For parameters near the Planck best fit, the run must reproduce reference CMB spectra within the numerical tolerance of the backend, produce positive physical densities, and satisfy flat-Friedmann closure. The validation tests are:

1. Compare C_ℓ^{TT}, C_ℓ^{TE}, C_ℓ^{EE}, and $C_L^{\phi\phi}$ against a reference CLASS/CAMB or Planck benchmark output.
2. Verify that $E^2(z)$ agrees with the sum of normalized energy densities and that $|\Omega_{\rm tot} - 1|$ is below numerical tolerance for a flat model.
3. Check $D_L(z) = (1+z)^2 D_A(z)$ and $D_M(z) = (1+z)D_A(z)$ in flat geometry.
4. Require positive D_A, D_L, $H(z)$, ρ_b, ρ_c, ρ_γ, and non-negative neutrino densities.
5. Confirm that $\sigma_8(z)$ and $f\sigma_8(z)$ behave physically over the redshift range used by the RSD data.

The validation archive should store plots of CMB spectra and residuals, $H(z)$ and distances, growth functions, a Friedmann-residual table, and a machine-readable pass/fail summary including software versions and numerical tolerances.

8. Post-processing and minimum outputs

The converged chains should be processed to obtain posterior means, medians, best-fit or maximum-posterior points, and 68% and 95% credible intervals. The minimum derived parameters are Ω_m, H_0, σ_8, $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$, r_d, and the age of the Universe. The minimum figure set is: CMB spectra and residuals; BAO distance ratios versus redshift; the SNe Hubble diagram and residuals; RSD $f\sigma_8(z)$ with the best-fit curve and posterior band; and a triangle plot for the primary and key derived parameters. The minimum table set is: primary-parameter constraints, derived-parameter constraints for each dataset combination, and the minimum χ^2 contribution by dataset.

9. Paper-level documentation

A paper or arXiv presentation should contain the following sections. The *Data and Methods* section defines the ΛCDM model, priors, Planck 2018 likelihood, BAO compilation, Pantheon/Pantheon+ SNe likelihood, and RSD measurements. The *Analysis Pipeline* section describes the CLASS/CAMB backend, likelihood construction, posterior definition, sampler settings, and incremental dataset combinations. The *Results* section reports constraints for Planck, Planck+BAO, Planck+BAO+SNe, and Planck+BAO+SNe+RSD, including H_0, Ω_m, σ_8, S_8, r_d, and dataset-level goodness of fit. The *Validation and Systematics* section documents the relativistic/CMB validation gate, dataset-removal tests, prior-sensitivity checks, BAO/RSD correlation treatment, SNe calibration choices, Planck foreground robustness, and optional CLASS-versus-CAMB benchmark comparisons.

For wCDM, the pipeline is generalized by adding w_0 with a broad prior such as $w_0 \sim U(-2, -0.3)$ and activating the dark-energy fluid sector in the Boltzmann solver. For w_0w_aCDM one uses $w(a) = w_0 + w_a(1-a)$ and validates the perturbation treatment, especially near phantom crossing. For interacting-vacuum or DVCH-like extensions, the background and perturbation equations must be implemented directly in CLASS/CAMB and the full validation gate must be rerun before any Planck+BAO+SNe+RSD posterior is interpreted.

10. Executed status of the CMB+late-time confrontation

The executed status of the data confrontation is as follows. The workspace contains the DVCH Planck YAML target, the DVCH Boltzmann source-table interface, the local modified matter-sector solver, and the late-time MCMC diagnostics. In WSL2 Ubuntu, the DVCH CAMB fork was compiled with `gfortran`; a smoke test produced finite scalar spectra with `DVCH_flag=True` and a distinct ΛCDM comparison. The Planck `clik` source produced a resolved shared library. Building on that toolchain, the two analyses reported in Secs. IX B–IX E were executed:

- the real-data late-time analysis (Pantheon+ 1580 SNe with $z > 0.01$ and no calibrators, DESI DR1 BAO in its published correlated form, 31 cosmic chronometers, r_d marginalised) with MCMC posteriors and nested-sampling evidence;
- the Planck 2018 TTTEEE Cobaya runs with the DVCH-patched CAMB: an MCMC stopped at $R-1 = 0.0295$ over 1.17×10^5 accepted samples in 5 chains (low-ℓ TT/EE and lensing blocks summed in), and the re-optimised nested minima with the official free nuisance sector ($\chi^2_{\rm min} = 2766.81$ for ΛCDM versus 2766.10 for DVCH, $n = 0.01$ at the prior edge, $p \simeq 0.70$). The background of the patched CAMB is identical to ΛCDM, so these are perturbation-sector numbers: *provisional*, see Sec. IX D.

The short integration chains in the archive remain only as integration-validation artifacts, superseded by the runs above.

The remaining open items are therefore: patching the CAMB *background* sector to integrate the CDM–vacuum conservation equation (without it there is no CMB comparison of the closure), meeting the $R-1 < 0.01$ criterion

with the free nuisance sector and distributing the chains, propagating the DVCH interaction into the photon–baryon hierarchy, and running the joint CMB+late-time posterior together with the growth (RSD) sector.

The supplied Planck likelihood code and baseline likelihood archive were inspected in the external workspace. The WSL2 build now provides the required C/Fortran, BLAS/LAPACK, CFITSIO, and MPI toolchain, and produces `libclik.so`. CAMB 2.0.4 and Cobaya 3.6.2 were verified locally, including a baseline CAMB calculation with $\sigma_8 = 0.80975$ at $H_0 = 67\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and a compiled DVCH spectrum smoke test. The automatically generated broad nuisance ranges are deliberately not used: they caused a zero-acceptance chain and are not official Planck priors.

Perspectiva futura: Extender DVCH a escalas cuánticas para derivar formalmente el acoplamiento entre la modificación local de ρ_Λ predicha por la cerradura y la propagación de una partícula de laboratorio (el parámetro ζ de la Sec. IX A), de modo que la cota $|\zeta| < 2.3 \times 10^{-7}$ pueda convertirse en una restricción directa sobre la teoría en lugar de sobre un acoplamiento aún no derivado.

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